Benford's Law and Regression Results Published in Articles in *New Zealand Economic Papers*

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Abstract

Benford's Law determines the way in which the digits that make up a 'legitimate' multi-digit number are distributed. A vast multi-disciplinary literature has illustrated the relevance of this law, and its usefulness in detecting fraudulent reporting. There is a small literature that investigates whether published regression results satisfy Benford's Law. A handful of these studies relate to such results in economics journals, and they provide evidence that this law is satisfied for legitimate results in this case. This paper extends that literature through formal statistical testing for Benford's 1st and 2nd digit distributions in all of the regression results reported in *New Zealand Economic Papers*, from its inception in 1966 to 2025. Both Bayesian and frequentist tests are used, with special attention being given to the issues associated with the extremely large sample sizes. Overall, the results provide substantial additional support for adherence to Benford's Law in this particular type of data.

JEL codes: **C11, C12, C46**

Key words: **Benford's Law, regression results, economic data**

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1. Introduction

Regression analysis is a common statistical tool that has been widely used in applied economic research for decades. Accordingly, the numerical results of this analysis provide the foundation for vast numbers of research papers published in numerous economics journals. Indeed, the actual numbers that constitute the regression results can be one of the key factors determining whether or not such papers are accepted for publication. In recent years there has been an increasing trend towards requiring that authors make the data associated with their empirical results available to journal referees and readers of their papers. Undoubtedly, this is a positive requirement, and it is consistent with this statement from one of the founding parents of econometrics: 'In statistical and other numerical work presented in *Econometrica* the original raw data will, as a rule, be published, unless their volume is excessive. This is important in order to stimulate criticism, control, and further studies.' (Frisch, 1933, p.3).

However, there are still many factors that may contribute to the publication of questionable results. For example, publicly available data may have been selected in a biased manner, with the intention of obtaining desired results. If the data come from surveys conducted by the author(s), the quality of the survey design may be unknown, and this may distort the results that are reported. For example, see Lovell (1983) and Denton (1985). In addition to providing access to the data that have been used, clear information about the econometrics packages that were used (including version numbers) and the provision of any software code (*e.g.*, R or Python), is also important if results are to be reproducible. Dewald *et al*. (1986) provide an excellent discussion and analysis that is totally pertinent today.

There may also be issues surrounding the selection of the reported results, relative to all of the results that were obtained during the empirical study. Historically, and in many fields, statistically significant results have been more likely to be acceptable than insignificant ones when making decisions regarding publication. In a 'publish or persish' environment, this can affect the extent to which published regression results are truly representative of the underlying facts, or of the research outcome at large. A recent example of this is provided by Brodeur *et al*. (2016), who analyzed the distribution of over 50,000 p-values from 641 empirical articles published in the *American Economic Review*, the *Journal of Political Economy*, and the *Quarterly Journal of Economics* between 2005 and 2011. They found that some 30,000 of the results imply rejection at the 10% significance level, 27,000 at the 5%, level, and 21, 000 at the 1% level. These figures align closely with statistically significant outcomes. In addition, they found that there is a 'shortage' of p-values between 0.25 and 0.10; and an excessive number slightly below 0.05.

This is part of a general issue has been the subject of considerable attention from researchers in many fields. Indeed, in some disciplines there have been suggestions that the publication of inappropriate empirical results is rife. For example, Bastardi *et al.* (2011) and Simmons *et al*. (2011) draw attention to such behaviour in psychology; and Horton *et al*. (2020) refer to similar activity in accounting and management research. One important response to this is to encourage the preparation and publication of replication studies. As McCullough and Vinod (2003, p.888) state: 'Replication is the cornerstone of science. Research that cannot be replicated is not science, and cannot be trusted either as part of the profession's accumulated body of knowledge or as a basis for policy.'

Historically, however, the number of research journals in economics (and most other social sciences) that adopted policies that encouraged and facilitated replication was quite limited. Interestingly, some key replication studies came to disturbing conclusions. For example, McCullough *et al.* (2006) were able to replicate the results of only 14 of 186 empirical articles published in the *Journal of Money, Credit, and Banking* during 1996-2003. McCullough (2007, p.335) concluded that: 'All available evidence indicates that replicable economic research is the exception and not the rule, ….'. More recently, there has been an increase in the publication of replication studies relating to economic research. Again, though, some of the results have been disturbing. For example, in their attempted replication of the results of 18 laboratory experimental studies published in the *American Economic Review* and the *Quarterly Journal of Economics* between 2011 and 2014, Camerer *et al*. (2016) were able to produce results that were significantly in accord with those published in only 61% of studies.

Partly in response to Dewald *et al*. (1986), the *American Economic Review* implemented a policy requiring authors to provide the information needed for the replication of their empirical results. Other economics journals that have similar (but in some cases weaker) policies include the *Journal of International Economics*, *Journal of Human Resources*, *International Journal of Industrial Organization*, *Empirical Economics*, the *Economic Record*, the J*ournal of Applied Economics*, the *Journal of Econometrics*, the *Journal of Business and Economic Statistics, Quantitative Economics,* the *Review of Economics and Statistics*, and *Econometrica*. *New Zealand Economic Papers* now follows their publishing house's 'data sharing policy', which states: 'Authors are encouraged to share or make open the data supporting the results or analyses presented in their paper where this does not violate the protection of human subjects or other valid privacy or security concerns.' In spite of such policies many economics journals choose not to actually publish replication studies themselves, although some notable exceptions include *Empirical Economics*, and the *Journal of Applied Econometrics*. On the other hand, the publication of replication studies is one of the primary objectives of the *Journal of Comments and Replications in Economics*, which superseded the former *International Journal for Re-Views in Empirical Economics* in 2022.

In this paper we analyze all of the reported regression results that have appeared in *New Zealand Economic Papers* (*NZEP*) since its inception in 1966 to 2025. This does not involve any attempts to replicate the results. Indeed, given the absence of data in the majority of cases, this would be impossible at this point in time. Instead, we undertake a statistical analysis of the actual numbers that constitute the reported results, with a view to determining if they satisfy a well-known form. To be clear, this is not to suggest that the validity of the multitude of regression results that have been published in *NZEP* since its very first issue should be questioned. To the best of our knowledge, no 'negative' replications of the associated papers have been published. Moreover, no retractions of any of these papers have been announced in the journal. This is very positive, and the 'data sharing policy' that is now in place is to be commended.

Our statistical analysis of the actual numbers that are reported for estimated regression coefficients and their standard errors and Student t-statistics is designed to strengthen the limited evidence that such numbers are consistent with 'Benford's Law' (Benford, 1938; and Newcomb, 1881). This 'law' states what statistical distribution the first significant digits of a collection of multi-digit numbers should follow. It also provides the expected distribution for the second and higher significant digits, and their joint distributions. The question to be addressed is: 'do the key numbers associated with regression results based on economic data generally follow Benford's Law?' If they do, then this provides a basis for identifying groups of such results whose validity is questionable because they fail to meet this requirement. In short, it provides a 'net' for catching questionable regression results, in situations where the lack of background data and/or software information precludes rigid replication studies. In addition, where such information is in fact available, a failure to meet Benford's Law may provide a flag as to which studies warrant (costly) attention.

As we discuss in the next section, Benford's Law has a long history as a tool to assist with data fraud, in many fields. However, in the specific case of regression results the evidence to date is surprisingly limited, and relatively recent. The early study by Diekmann (2007) assessed tabulated results for OLS regression coefficients published in the January 1996 and May 1997 issues of the *American Journal of Sociology*. The first significant digits of these numbers were found to closely approximate Benford's Law. The second digits of corresponding estimates in a second sample from two later volumes of the same journal also aligned closely with Benford's Law. Bauer and Gross (2011) produced similar results with sociological data, as did Auspurg and Hinz (2017) for regression test statistics from social psychological studies. However, for published regression results based on economic data, the only studies that assess Benford's Law are those of Günnel and Tödter (2009), Tödter (2009), Brodeur *et al.* (2016), and more recently Lazebnik and Girlitsky (2023). In each case, the authors find a very strong alignment with Benford's Law for the first (and in some cases second) significant digits of coefficient estimates and standard errors. The present paper extends these studies by using a large set of data that covers regression results associated with many areas of empirical economics. Our results are consistent with the counterparts noted above.

The rest of the paper is structured as follows. The next section provides a brief overview of Benford's Law and discusses some of its applications. Section 3 describes the data that have been gathered for this study; and the statistical methods that are used to assess the data are discussed and justified in Section 4. In section 5 we present the results of our analysis, based on both the full set of data and various sub-samples. Some concluding remarks, as well as suggestions for further related research, appear in the final section.

2. Benford's law

The result that is usually referred to as Benford's Law emerged from an observation by Newcomb (1881) that tables for logarithms of numbers with small first digits showed more wear (and hence more use) than those of numbers with larger first digits. This point was 're-discovered' by Benford (1938), as a result of studying over 20,000 numbers from a wide range of fields. The formal mathematical basis for Benford's Law is provided by Berger and Hill (2015, 2020). Here we simply present a brief overview of some key aspects of this law, expressed in terms of statistical distributions.

Let *X* be a (real) random variable, with *x* being a value that it takes. Let $D_j(x)$ denote the *j*'th. significant digit of *x* (for *j* = 1, 2, …). For example, if *x* = 2.3016, then $D_1(x) = 2, D_2(x) = 3, D_3(x) = 0, D_4(x) = 1, D_5(x) = 6$. Similarly, if *x* = - 0.0023, then $D_1(x) = 2, D_2(x) = 3$. In general, we will denote the *value* of the *j*'th. significant digit of *x* by d_j, where $d_1 \in [1, 9]$ and $d_s \in [0, 9]$ for all $s > 1$.

If *X* follows Benford's distribution (*i.e.*, it satisfies Benford's Law), then the following result holds, as proven by Hill (1995a):

$$Pr.[D_1(x) = d_1, D_2(x) = d_2, \dots\dots\dots, d_m(x) = d_m] = log\left[1 + \left(\textstyle\sum_{i=1}^{m} 10^{m-i} d_i\right)^{-1}\right] \quad . \qquad (1)$$

The logarithms here (and in all of our related discussion) have a base of 10. This aligns with the decimal basis of our number system. So, for example, if *m* = 2 in (1), then

$$Pr.[D_1(x) = d_1, D_2(x) = d_2] = log\left[1 + \frac{1}{(10d_1 + d_2)}\right]$$

$$= \log[10d_1 + d_2 + 1] - \log[10d_1 + d_2] \quad . \qquad (2)$$

These particular joint probabilities are shown in Table 1. Summing across the columns (rows) in that table yields the marginal distribution for d_1 (d_2). In addition, summing the expression in (2) over the values for $d_2 \in [0, 9]$ yields the marginal Benford distribution for the first significant digit, d_1:

$$\begin{aligned} Pr.(D_1(x) = d_1) &= \log[10d_1 + 1] - \log[10d_1] + \log[10d_1 + 2] - \log[10d_1 + 1] \\ &\quad + \cdots \ldots \ldots . + \log[10d_1 + 9] - \log[10d_1 + 8] \\ &\quad + \log[10d_1 + 10] - \log[10d_1 + 9] \\ &= \log[10d_1 + 10] - \log[10d_1] \\ &= \log[1 + 1/d_1]\ ; \qquad \text{for } d_1 \in [1, 9] \end{aligned} \tag{3}$$

The particular result in equation (3) for the first significant digit is what most people think of as Benford's Law. This marginal discrete probability distribution is depicted in Figure 1, along with its counterparts for the second and third significant digits. The marginal distribution for d_2 can be computed by summing the expression in (2) over $d_1 \in [1, 9]$. That for d_3 can be obtained by taking the joint distribution for d_1, d_2, and d_3 from (1), by summing over $d_1 \in [1, 9]$ and $d_2 \in [0, 9]$. The values for the marginal Benford distributions of the first four significant digits appear in Table 2.

The means of the Benford (marginal) distributions for the first four significant digits are 3.4402, 4.1874, 4.4678, and 4.4967 respectively. The corresponding variances are 6.0565, 8.2538, 8.2501, and 8.2500. Given the values of $p(d_3)$ and $p(d_4)$ in Table 2, and noting that the exact mean and variance of a discrete random variable that follows a uniform distribution on [0 , 9] are 4.5 and 8.25 respectively, it is clear that Benford's distribution for d_k converges quickly to a U[0 , 9] distribution for $k \geq 3$.

Table 1: Benford's distribution – joint probabilities, $p(d_1 , d_2)$

d_1/d_2	0	1	2	3	4	5	6	7	8	9	$p(d_1)$
1	0.0414	0.0378	0.0348	0.0322	0.0300	0.0280	0.0263	0.0248	0.0235	0.0223	**0.3010**
2	0.0212	0.0202	0.0193	0.0185	0.0177	0.0170	0.0164	0.0158	0.0152	0.0147	**0.1761**
3	0.0142	0.0138	0.0134	0.0130	0.0126	0.0122	-.0119	0.0116	0.0113	0.0110	**0.1249**
4	0.0107	0.0105	0.0102	0.0100	0.0098	0.0095	0.0093	0.0091	0.0090	0.0088	**0.0969**
5	0.0086	0.0084	0.0083	0.0081	0.0080	0.0078	0.0077	0.0076	0.0074	0.0073	**0.0792**
6	0.0072	0.0071	0.0069	0.0068	0.0067	0.0066	0.0065	0.0064	0.0063	0.0062	**0.0669**
7	0.0062	0.0061	0.0060	0.0059	0.0058	0.0058	0.0057	0.0056	0.0055	0.0055	**0.0580**
8	0.0054	0.0053	0.0053	0.0052	0.0051	0.0051	0.0050	0.050	0.0049	0.0049	**0.0512**
9	0.0048	0.0047	0.0047	0.0046	0.0046	0.0045	0.0045	0.0045	0.0044	0.0044	**0.0458**
$p(d_2)$	**0.1197**	**0.1139**	**0.1088**	**0.1043**	**0.1003**	**0.0967**	**0.0934**	**0.0904**	**0.0876**	**0.0850**	**1.0000**

Figure 1:
Benford's Distributions for 1st. to 3rd. Digits

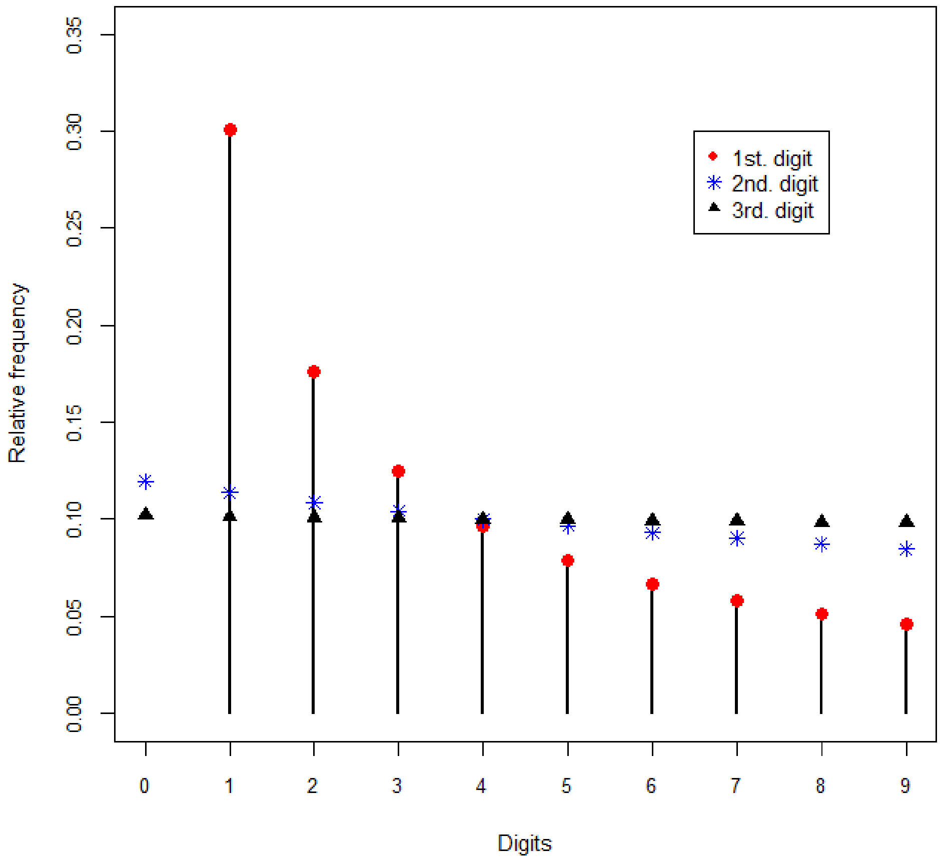


Table 2: Benford's marginal distributions - first four significant digits

***d*$_i$:**	**0**	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**
$p(d_1)$	n.a.	0.3010	0.1761	0.1249	0.0969	0.0792	0.0669	0.0580	0.0512	0.0458
$p(d_2)$	0.1197	0.1139	0.1088	0.1043	0.1003	0.0967	0.0934	0.0904	0.0876	0.0850
$p(d_3)$	0.1018	0.1014	0.1010	0.1006	0.1002	0.0998	0.0994	0.0990	0.0986	0.0983
$p(d_4)$	0.1002	0.1001	0.1001	0.1001	0.1000	0.1000	0.0999	0.0999	0.0999	0.0998

Benford distributions satisfy several properties that are important in many applications. First, Benford (1938) established that they are inversion invariant. If the data satisfy a Benford distribution, so do the inverses of the data. Second, they are sum invariant (Nigrini, 1992; Allaart, 1997). Third, as was proven by Pinkham (1961), they are invariant to a change of scale in the data. So, if Benford's law holds for a class of data measured in N.Z. dollars, it will still hold if the data are measured instead in Japanese yen (at a fixed exchange rate). Moreover, Pinkham proved that Benford's distribution is the only possible distribution for first significant digits that is scale invariant. Fourth, Benford's Law is invariant to the choice of base for the digits (Hill, 1995b). So, if the numbers are duodecimal (rather than base 10), and base 12 logarithms are used, then equations (1) to (3) still apply.

Hill (1995a, 1995b) presented rigorous proofs of these invariance results, and he also established a statistical limit law:

> 'Roughly speaking, this law says that if probability distributions are selected at random and random samples are then taken from each of these distributions in any way so that the overall process is scale (or base) neutral, then the significant-digit frequencies of the combined sample will converge to the logarithmic distribution.' (Hill, 1995a, p.360).[2]

This last result provides a justification for assessing the quality of reported data by using Benford's distribution as a benchmark. That is: (his) 'Theorem 3 suggests a straightforward test for unbiasedness of data – simply test goodness-of-fit to the logarithmic distribution. (Hill, 1995a, p.361). It also helps to explain why Benford's Law emerges in a very wide range of data types, data contexts, and fields of study.

Indeed, the literature associated with tests of Benford's Law is vast.[3] It includes empirical studies that cover data from all corners of the physical, medical, and social sciences. Berger and Hill (2015) and Miller (2015) provide detailed discussions of this literature. Some examples of studies that illustrate the range of associated fields of application include: pollen counts (Docampo *et al*., 2009), electoral data (Pericchi and Torres, 2011), population sizes (Jamain, 2001), eBay auctions (Giles, 2007), genome sizes (Friar *et al.*, 2012), image compression and reconstruction (Jolion, 2002; Sanches and Marques, 2006), computer science (Safaei *et al*., 2022), stock prices (Ley, 1996; Pietronero *et al*., 2001; Shengmin and Wenchao, 2010), classical music (Khosravani and Rasinariu, 2018), precipitation data (Dollop *et al*., 2026), and income tax data (Nigrini, 1992, 1996, 2012). The last of these authors popularized the use of Benford's Law in forensic studies – specifically in the context of accounting data, and especially taxation data. Subsequently, Benford's Law assisted many attempts to expose fraudulent data. Some justification for this comes from the results of Horton *et al.* (2020) and Lazebnik and Girlitsky (2023) who compared legitimate data with knowingly corrupted data, and found that Benford's Law helps to identify the latter.

Interestingly, the literature associated with using Benford's Law to assess the legitimacy of the results of statistical analysis in published research papers is rather limited, as we summarized in section 1. Only four papers relate to regression results based on economic data. This is our focus – specifically, reported regression results in the form of coefficient estimates, standard errors, and Student t-statistics. The limited evidence to date suggests that these data should follow Benford's Law if calculated and reported correctly. Indeed, there is some foundational support for this from the results of Shengmin and Fasli (2001), who found that if the numbers in a set of data are formed as products of a sufficiently large number of terms, then Benford's 1^{st} - digit distribution will be satisfied. Such multiplication underlies standard regression analysis.

3. Data

The data used in this study are based on all of the regression results reported in every issue of *NZEP* between Volume 1 (Issue 1) in 1966 and Volume 59 (Issue 1) in 2025. There was no publication in 1967, so Volume 2 appeared in 1969. The data cover all regression results presented in tables, passages of the text, and appendices. A wide range of regression models is included over the years, such as single equation static and dynamic models, simultaneous equations models, error-correction models, limited dependent variable models, *etc*. Accordingly various estimation methods are used by authors to obtain their results, including ordinary least squares, instrumental variables, various maximum likelihood estimators, and others. In addition, the data that are used for the regression models relate to a wide range of branches of economics, and come from numerous sources.

The number of papers that include regression results varies, of course, from Issue to Issue (and Volume to Volume) of the journal, as is shown in Figure 2. There is a clear upward trend in the number of these papers per Volume since *NZEP* was first published, especially since 1984 (Volume 18). A total of 280 papers with regression results were published during the period under study, and the number of such papers per Volume varied from zero (in 1982 and 1984) to 14 (in 2024). It also ranged from 0% to 77.8% of the *total* number of papers per Volume (*i.e.*, per year), with a median and mean of 34% and 36% respectively. Within these bounds, the number of estimation results per analyzed per paper also varies substantially. All original articles and notes, as well any corrigenda were considered when counting the total number of papers in a Volume. However, other published pieces such as editors' introductions, review papers, and announcements were not included in the 'total number of papers per Volume' count.

The specific data that we analyze in this study comprise the coefficient estimates, their standard errors, and Student t-statistics (or, equivalently, Z statistics for large-sample situations). All reported coefficient estimates are analyzed. In some cases, neither standard errors nor t-statistics are reported. If p-values are given, these are ignored. If both standard errors and t-statistics are presented for a particular case, then only the standard errors are analyzed. We do not use the coefficient estimates and t-statistics to impute standard errors as this can easily result in rounding errors, which can negatively impact the crucial significant digits and undermine Benford's Law. Consequently, over the full data-set there are more estimated coefficients than the sum of the number of reported standard errors and Student t-statistics; and the numbers of each of these vary considerably from year to year. This is illustrated in Figure 3. In addition, some of the reported results contain only one significant digit, so sample sizes for 2nd - digit analysis are smaller than those for 1st - digit analysis. The full sample of data that is analyzed is summarized as in Table 3, where values 'per paper' refer to the papers that are analyzed because they contain regression results.

Figure 2:
Number of Regression Papers per Volume

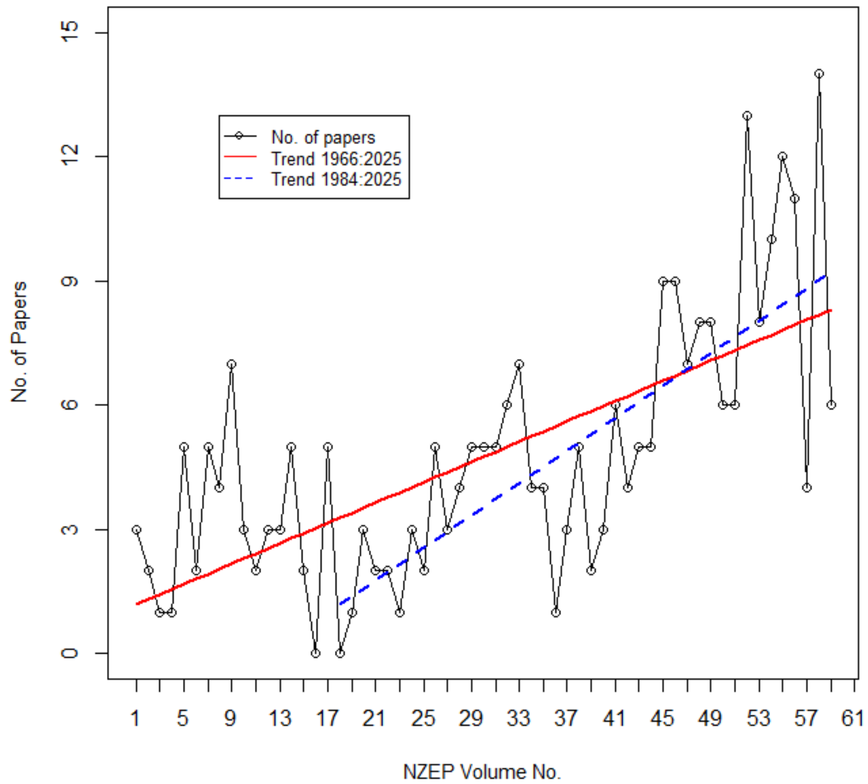


Figure 3:
Number of Coefficients, Standard Errors, & t-Statistics per Volume

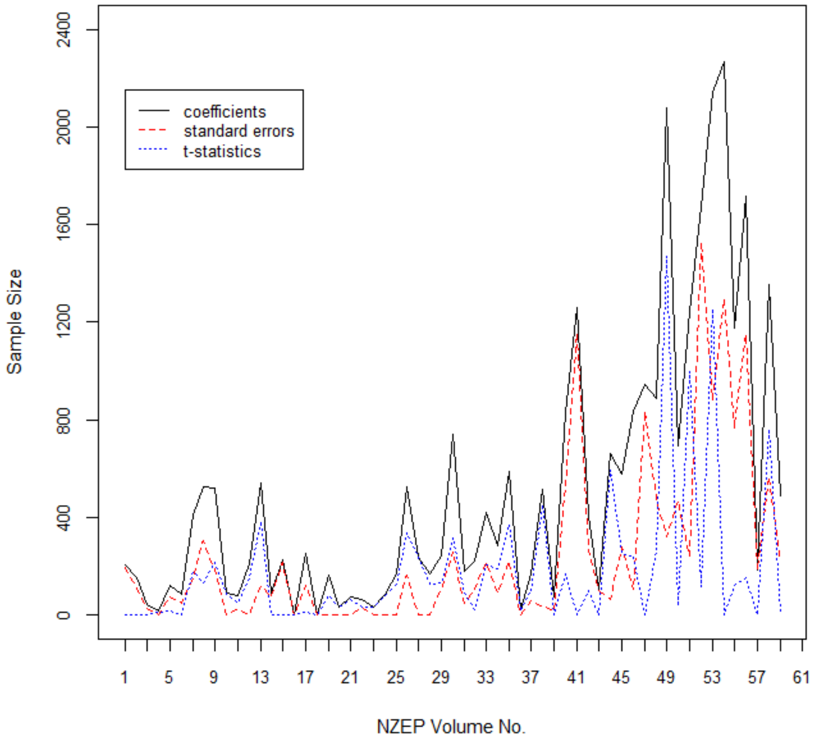


Table 3: Full sample summary

	Coefficients	Std. errors	Student - t statistics
No. of 1st. digits	30,159	14,519	10,841
No. of 2nd digits	27,379	12,885	10,605
Min. No. per Volume	0	0	0
Max. No. per Volume	2,267	1,527	1,468
Mean No. per Volume	511.2	246.1	183.7
Median No. per Volume	241.0	109.0	92.0
Std. Dev. Per Volume	570.0	351.8	294.0
Min. No. per paper	11.7	0	0
Max. No. per paper	277.0	192.3	183.5
Mean No. per paper	92.8	40.8	37.0
Median No. per paper	77.5	26.3	26.8
Std. Dev. per paper	66.8	44.6	44.0

4. Statistical methodology

4.1 Background

Traditionally, researchers have used graphical displays and various ‘goodness-of-fit’ (GOF) tests to try and determine the applicability of Benford’s Law to different data. For those tests, the null hypothesis is that the data follow a Benford distribution. In the case of first significant digits, that distribution is:

$$Pr.(D_1(x) = d_1) = \log[1 + 1/d_1] \quad ; \quad \text{for } d_1 \in [1, 9] \quad , \tag{4}$$

and in the case of the second significant digit it is:

$$Pr.(D_2(x) = d_2) = \textstyle\sum_{k=1}^{9} log[1 + 1/(10k + d_2)] \quad ; \quad \text{for } d_2 \in [0, 9] \quad . \tag{5}$$

A two-sided alternative (‘not Benford’) hypothesis applies to those tests, which include (among others), the chi-square test and various empirical distribution function (EDF) tests such as the Anderson-Darling, Cramér-von Mises, Watson, and Kuiper tests. Other tests that have been used have been based on the mean absolute deviation (MAD) between the Benford probabilities and the sample relative frequencies of the significant digits. Variations on MAD-based tests include those proposed by Cerqueti and Lupi (2022) and by Kösssler *et al*. (2024, and 2026), and the latter authors also introduce ‘invariant sum’ tests for the null hypotheses in (4) and (5). Tödter (2009) proposes a test based on a weighted average of the nine (ten) digits associated with d_1 (d_2), and applies it to regression results data in certain issues of *Empirica* and *Applied Economics Letters*.

While the majority of these (and other related) tests have well-established asymptotic distributions, their use with extremely large samples is questionable. As with any asymptotically justified tests, they have exceptionally high power when the sample is very large, but this can be problematic, as has been discussed widely since Berkson (1938) and Lindley (1957). As Campanelli (2022, p.54) notes: ‘In particular, they have enormous power for large *N* making them too rigid to assess the goodness-of-fit well: even a tiny deviation of the 1st - digit counts from Benford’s distribution will be statistically significant.’ Other authors have also recognized this problem. Miller (2015) states that, ‘It is a non-trivial task to find good statistical tests for large data sets’. This issue has been discussed broadly in the context of testing for Benford’s Law by Kossovsky (2021), Cerqueti and Lupi (2022), among others. The sample sizes used in this paper are relatively large, and so the choice of statistical tests is crucial[4].

Another factor that is often overlooked is that most standard GOF tests are based on the assumption that the data are continuous. When testing Benford’s Law the data are discrete digits, and special care must be taken over the construction of GOF tests to ensure that they remain distribution-free, and that the correct critical values are used. Various modifications of EDF tests have been proposed, including those by Freedman (1981), Choulakian *et al.* (1994), and Lockhart *et al.* (2007), for example. Giles (2013) derived *exact* asymptotic critical values for Freedman's test for Benford distributions; and Lesperance *et al*. (2016) derived approximate critical values for the tests of Lockhart *et al.* in this context.

All of the tests mentioned so far are based on frequentist statistical inference. In the same vein, the visual assessment of plots, that compare Benford's distribution with sample relative frequencies for significant digits, can be augmented with confidence intervals. Again, care must be taken over the construction of such intervals. As Lesperance *et al*. (2106) observe, in this context confidence intervals based on individual digit values are distorted, and their confidence levels biased. Simultaneous (multinomial) confidence intervals based on the full set of (nine or ten) digits are required, and they recommend Goodman's (1965) formulation.

Of course, as an alternative to frequentist inference, Bayesian inference can be adopted to test the hypothesis that Benford's Law is satisfied. Specifically, formal testing of the hypotheses in (4) and (5) can be conducted using 'Bayes factors' (or full 'Bayesian posterior odds'), the merits of which are discussed by Kass and Raftery (1995), for example. This approach was adopted to test for Benford's Law by Pericci and Torres (2011) and Teles da Fonesca (2016), for example, using both compound multinomial-Dirichlet, and compound binomial-beta models. As with any Bayesian inference, the sensitivity of the results to the choice of prior distributions needs to be reported. However, a crucial advantage of the use of Bayes factors is that it avoids the issues associated with very large samples that was discussed above, as Kass and Raftery (1995, p.789) state: 'Frequentist tests tend to reject null hypotheses almost systematically in very large samples, whereas Bayes factors do not.'

4.2 Bayesian inference

We use both frequentist and Bayesian methods in this study. This provides a comprehensive set of results, and illustrates the pros and cons of each approach to testing Benford's Law. Our Bayesian analysis involves the Bayes factors associated with the multinomial-Dirichlet model, for both 1st - digit and 2nd - digit cases. The null hypothesis, H_0, is given by (4) or (5), and the alternative hypothesis, H_1, is that the digits do not follow that Benford distribution. The prior probabilities for these hypotheses are $p(H_0) = \pi_0$ and $p(H_1) = 1 - \pi_0$, so the prior odds in favour of H_0 are $O_{01} = \pi_0/(1 - \pi_0)$.

The Bayes factor in favour of H_0 over H_1 is based on the vector, $x = (x_1, x_2, \dots, x_k)$, whose elements are the 'digit counts' for $k = 9$ (10) when testing first (second) significant digits. The joint density of x is:

$$f(x|\theta) = n! \prod_{i=1}^{k} \theta_i^{x_i} / \prod_{i=1}^{k} x_i! \qquad (6)$$

where $\theta = (\theta_1, \theta_2, \ldots., \theta_k)$ is the vector of parameters. If H_0 is true, then $\theta = \theta_0 = (p_1, p_2, \ldots., p_k)$, where p_i is the probability that $d_1 = i$, from (4) for the 1st - digit case; or the probability that $d_2 = i$, from (5) for the 2nd - digit case. The conjugate prior distribution for the parameter vector is Dirichlet, with density function:

$$f(\theta|\alpha, H_1) = \Gamma\left(\sum_{i=1}^{k} \alpha_i\right) \prod_{i=1}^{k} \theta_i^{\alpha_i - 1} / \prod_{i=1}^{k} \Gamma(\alpha_i)\ , \tag{7}$$

where $\alpha = (\alpha_1,\ \alpha_2, \ldots \ldots, \alpha_k)$, the parameter vector for the prior density for θ, can take various forms. To illustrate the sensitivity of our results to the choice of this prior, as is important in any Bayesian analysis, we consider four choices of α:

(i) $\alpha = \alpha_1 = (1, 1, \ldots \ldots, 1)$, in which case $f = (k-1)!$, which is 'uniform', as discussed by Pericchi and Torres (2011, p.507).

(ii) $\alpha = \alpha_2 = (k^{-1}, k^{-1}, \ldots.., k^{-1})$, in which case $f(\theta|\alpha, H_1)$ is the 'overall objective prior' for θ. See Berger *et al.* (2015).

(iii) $\alpha = \alpha_3 = (cp_1, cp_2, \ldots \ldots, cp_k\ )$, where $c = 22$ when H_0 corresponds to (4), and $c = 12$ when H_0 corresponds to (5). In this case $f(\theta|\alpha, H_1)$ is the 'least informative' natural conjugate prior for θ. See Teles da Fonesca (2016, pp. 20-23).

(iv) $\alpha = \alpha_4 = (p_1, p_2, \ldots \ldots, p_k\ )$. See Teles da Fonesca (2016, p.28).

Then, the Bayes factor in favour of H_0 over H_1 is the ratio of the *marginal* data density under H_0 to that under H_1. These can be shown to be:

$$m_0(x) = n! \prod_{i=1}^{k} {\theta_{0i}}^{x_i} / \prod_{i=1}^{k} x_i! \tag{8}$$

and

$$m_1(x) = n!\, B(\alpha + x) / \left[B(\alpha) \prod_{i=1}^{k} x_i!\right] \tag{9}$$

respectively, where $B(\ .\ )$ is the multivariate beta function. It then follows that the Bayes factor is

$$B_{01}(x) = \prod_{i=1}^{k} ({\theta_{0i}}^{x_i}) \prod_{i=1}^{k} [\Gamma(\alpha_i)]\, \Gamma\left[\sum_{i=1}^{k} (\alpha_i + x_i)\right] / \left\{\Gamma\left[\sum_{i=1}^{k} \alpha_i\right] \prod_{i=1}^{k} [\Gamma(\alpha_i + x_i)]\right\}\ . \tag{10}$$

By Bayes' Theorem, the Bayesian posterior odds (BPO) in favour of H_0 are $BPO_{01} = O_{01} B_{01}(x)$, and so the posterior probability that Benford's Law (H_0) is true is

$$p(H_0|x, \alpha\ ) = [1 + (1 - \pi_0)/(\pi_0 B_{01}(x))]^{-1}. \tag{11}$$

The digit counts that make up the elements of the x vector are extremely large in our analysis. Accordingly, we compute and report $\log_{10}[B_{01}(x)]$, together with $p(H_0|x, \alpha\ )$, based on equal prior odds. That is, with $\pi_0 = 0.5$. In this case, $p(H_0|x, \alpha\ ) = B_{01}(x)/[1 + B_{01}(x)]$.

4.3 Frequentist inference

In the following discussion, n is the sample size, f_i denotes the observed frequency of occurrence of the i^{th} value of the digit under test, and p_i denotes the corresponding hypothesized probability (determined by (4) or (5), and shown in Table 2). So, $i \in [1, 9]$ and $k = 9$ for the first significant digit, and $i \in [0, 9]$ and $k = 10$ for the second digit. We apply several frequentist tests in this study, despite their potential vulnerability to our large sample sizes.[4] The first, and most widely used, of these tests is the standard χ^2 GOF test, with the test statistic:

$$Q_{n,k} = n \sum_{i=1}^{k} \left[\frac{\left(\frac{f_i}{n} - p_i\right)^2}{p_i} \right] , \tag{12}$$

whose asymptotic distribution (under H_0) is $\chi^2_{(k-1)}$, where $(k - 1)$ is the degrees of freedom.

However, for very large sample sizes (and following earlier work by Lehmann and Romano, 2005), Cerquiti and Lupi (2022, p.13) determine that the null distribution of $Q_{n,k}$ is better approximated by a non-central χ^2 distribution with $(k - 1)$ degrees of freedom and a non-centrality parameter of $\lambda = (0.0238 \times n)$ in the 1st - digit case, and $\lambda = (0.0474 \times n)$ for the 2nd - digit test. This amendment has substantial implications for the first four moments of the null distribution, and the higher quantiles, as is illustrated in Table 4 for samples of size n = 1,000 and n = 10,000, and the 95th quantile, $q_{(0.95)}$. We see, for example, that for n = 10,000, the suggested 5% critical value when testing the 1st - digit null hypothesis (4) is 298.82 instead of 15.51.

Table 4: Non-central χ^2 features

	1st Digit			2nd Digit		
λ **(*n*)**	0	23.8 (1,000)	238.0 (10,000)	0	47.4 (1,000)	474 (10,000)
Mean	8.00	31.80	246.00	9.00	56.40	483.00
Variance	16.00	111.20	968.00	18.00	207.60	1950.00
Skewness	1.00	0.54	0.19	0.94	0.40	0.14
Kurtosis	4.50	3.40	3.05	4.33	3.22	3.03
$q_{(0.95)}$	15.51	50.63	298.82	16.92	81.65	556.62

The second test that we apply is that of Freedman (1981) – an EDF test modified for discrete data. Exact asymptotic critical values for the test, specific to the null hypotheses in (4) and (5) against a two-sided alternative, are provided in Table 2 of Giles (2013) These *exact* critical values were obtained by using the methodology[5] of Davies (1980). Freedman's test statistic is:

$$U_n^{*2} = (\tfrac{n}{k})\left[\textstyle\sum_{j=1}^{k-1} S_j^2 - \left(\sum_{j=1}^{k-1} S_j\right)^2/n\right], \tag{13}$$

where $S_j = \sum_{i=1}^{j}\left(\frac{f_i}{n} - p_i\right)$, for j = 1, …., k - 1, and the null hypothesis is rejected if U_n^{*2} exceeds the (asymptotic) critical value for the chosen significance level.

The mean absolute deviation between the k Benford probabilities, and their observed empirical relative frequency counterparts is another GOF measure that has long been a suggested basis for testing for Benford's Law. We use two MAD statistics in this paper. One is that employed by Nigrini (1992,1996, 2012):

$$MAD_N = (\tfrac{1}{k})\textstyle\sum_{j=1}^{k} |p_j - (f_j/n)|, \tag{14}$$

where 'N' identifies his name. He suggested (Nigrini, 2012, p.160) that a value of this statistic less than 0.015 would imply conformity of the data with Benford's 1st - digit law. Of course, this reasoning requires a formal statistical justification. Cerqueti and Lupi (2022, pp.6-7) establish that for large n, the distribution of MAD_N is approximately $N\left[\sqrt{2/(\pi n k^2)}\mathbf{1}'D\mathbf{1}; \mathbf{1}'DRD\mathbf{1}/(nk^2)\right]$. Here, $\mathbf{1}$ is a k-vector of ones and the $(k \times k)$ matrix D is defined as $D = diag.\sqrt{p_i(1-p_i)}$. The matrix $R = \{r_{ij}\}$ is $(k \times k)$ with elements

$$r_{ij} = \left(\tfrac{2}{\pi}\right)\left(\rho_{ij}\arcsin(\rho_{ij}) + \sqrt{1-\rho_{ij}^2}\right) - (2/\pi),$$

where $\rho_{ij} = -\sqrt{p_i p_j - (1-p_i)(1-p_j)}$; if $i \neq j$; and $\rho_{ii} = 1$.

Standardizing MAD_N using its large-n mean and standard deviation provides a standard-normal test statistic. Kösssler *et al*. (2024) suggest an alternative MAD statistic that we also employ:

$$MAD_K = \sqrt{n}\textstyle\sum_{j=1}^{k} |p_j - (f_j/n)|, \tag{15}$$

and they derive critical values (reported in their Table 3 and 4) with the aid of the large-sample distribution of a related statistic discussed by Cerqueti and Lupi (2021).

The fourth class of tests that we use for the null hypotheses in (4) and (5) are so-called invariant sum (IS) tests, proposed by Kösssler *et al*. (2024, 2026). These are motivated by the sum invariance property of Benford distributions, noted in section 2, and the results of Barabesi *et al*. (2020). The foundation for the test statistics is the normalized sum of the significands of the sample numbers, and asymptotic tests are constructed from the distance of these sums from zero. Two distance measures are used – Euclidean and Mahalanobis – resulting in test statistics denoted $IS_{d,E}$ and $IS_{d,M}$ respectively, with d = 1, 2 denoting tests of the first and second significant digits. Theorem 2 of by Kösssler *et al*. (2024) establishes that the asymptotic null distributions of $IS_{1,M}$ and $IS_{2,M}$ are χ^2, with 9 and 10 degrees of freedom respectively; and those of $IS_{1,E}$ and $IS_{2,E}$ are weighted sums of independent $\chi^2_{(1)}$ distributions, which can in turn be approximated by a linear transformation of a χ^2 distribution with 7.846 degrees of freedom for $IS_{1,E}$ and 9 degrees of freedom for $IS_{2,E}$. The details of the transformation are provided in their Appendix C.

We apply the *M* test suggested by Tödter (2009, p.342). This test is based on the statistic

$$M = n\left[\bar{d} - E(d)\right]/Var(d)\,, \tag{16}$$

where $\bar{d} = n^{-1}\sum_{i=1}^{k} f_i d_i$, and k = 9 (10) when testing the 1st (2nd) digit null hypothesis. The expectation and variance in the 1st digit case are 3.4402 and 6.0565 respectively, and in the 2nd digit case they are 4.1874 and 8.2538. By the central limit theorem, the asymptotic distribution of *M* is $\chi^2_{(1)}$ under either of the null hypotheses, (4) and (5). We adopt this test because Tödter applies it to data of the type under study in this paper, and because he commends this test for its relative power.

Finally, in some of our key plots that compare empirical relative frequencies with Benford distributions we include Goodman (1965) simultaneous (and asymptotic) confidence intervals based on the former statistics. For a sample size n, a $100(1-\alpha)\%$ Goodman confidence interval takes the form,

$$\left[B + 2f_i \mp \{B[B + 4f_i(n - f_i)/n]\}^{1/2}\right]/[2(n + B)] \quad , \tag{17}$$

where $B = \chi^2_{(1,\alpha/k)}$.

5. Results

All of the results were obtained using R code (R Core Team, 2026) written by the author.[6] We first present our results based on the full samples of data for regression coefficients, standard errors, and t-statistics, as summarized in Table 3. Descriptive results are followed by those based on the various frequentist tests. As discussed previously, the reliability of these tests is questionable for our large sample sizes. However, their widespread use requires that their results be included before focusing on the Bayesian test results that we prefer. The robustness of the basic results is then illustrated by presenting their counterparts for the sub-samples based on decades and on random selection. The last of these cases involves sample sizes that offer a stronger foundation for the frequentist tests.

5.1 Full sample results

We begin with a visual comparison of the empirical relative frequencies of the first and second digits with their Benford distribution counterparts in the first two rows of Table 2. These appear in Figures 4 and 5 respectively. Figure 4 suggests that the first significant digits of the coefficients, standard errors, and t-statistics all align well with Benford's Law. This may appear less clear in Figure 5, but the substantially different vertical axis scale (consistent with Figure 1) should be noted. These visual impressions require a statistical foundation, which is provided in Figures 6 to 8 with the addition of simultaneous 95% Goodman confidence intervals. These provide support for Benford's Law by covering the Benford probabilities in all cases except for the Student-t statistic first digit $d_1 = 2$, as is seen in Figure 8(a).[7] In Figure 6(a), the confidence interval boundaries for $d_1 = 2$ and $d_1 = 3$ coincide with the hypothesized probabilities, but the coverage is complete to three decimal places if a 97.5% confidence level is used. In short, these results provide extremely strong support for the consistency of the *NZEP* regression results with Benford's Law.

Figure 4:
Relative Frequency of 1st. Digits of Coefficients, Standard Errors, & t-Statistics

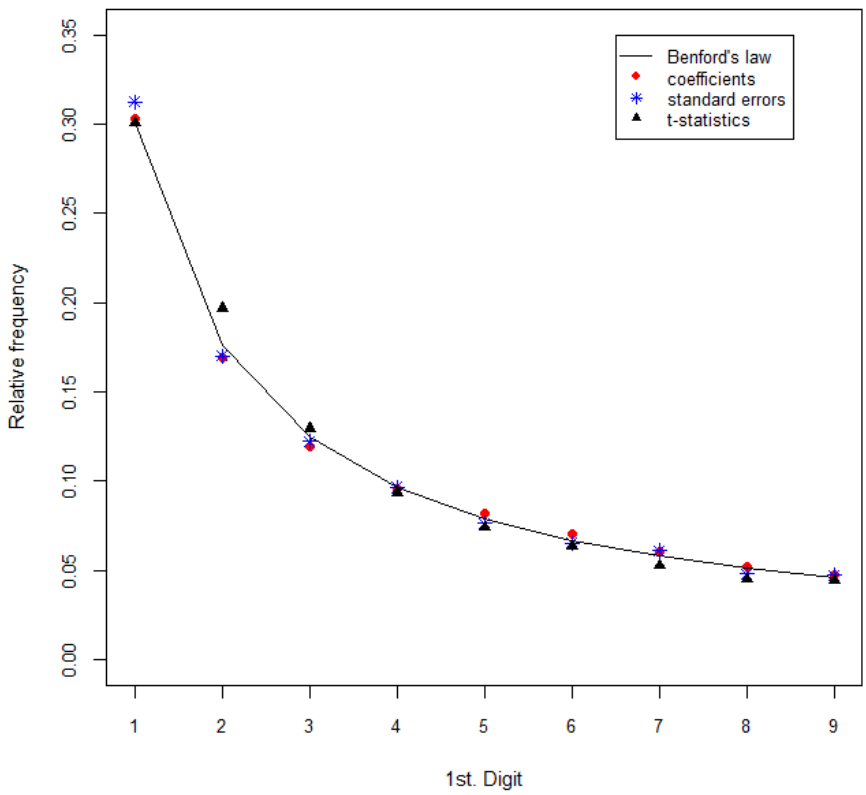


Figure 5:
Relative Frequency of 2nd. Digits of Coefficients,
Standard Errors, & t-Statistics

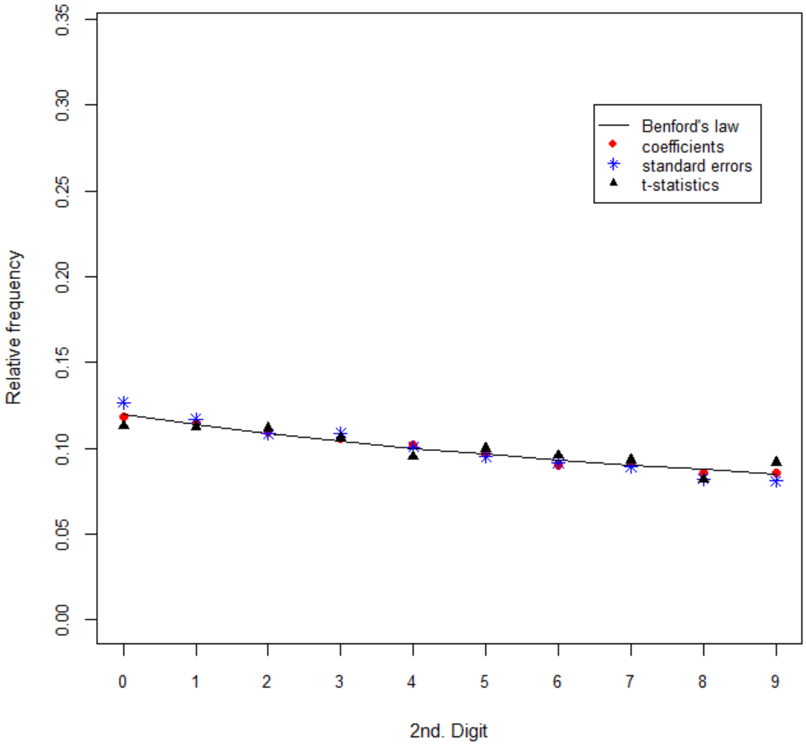


Figure 6(a): 95% Goodman Confidence Intervals
for Coefficient 1st. Digits

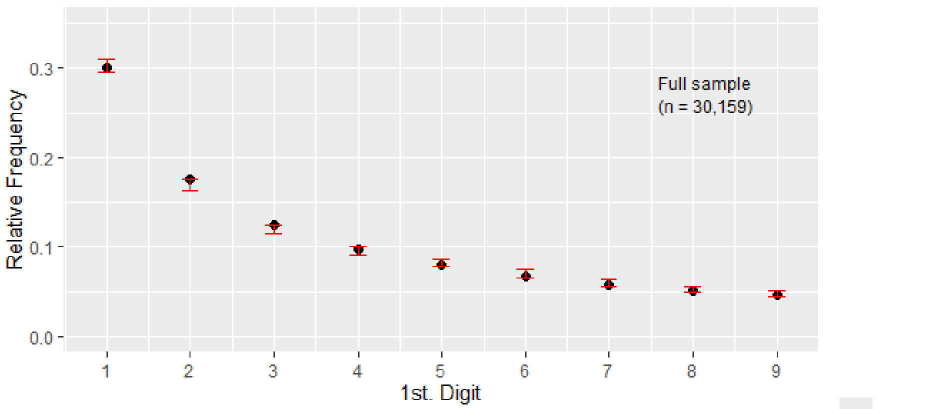


Figure 6(b): 95% Goodman Confidence Intervals
for Coefficient 2nd. Digits

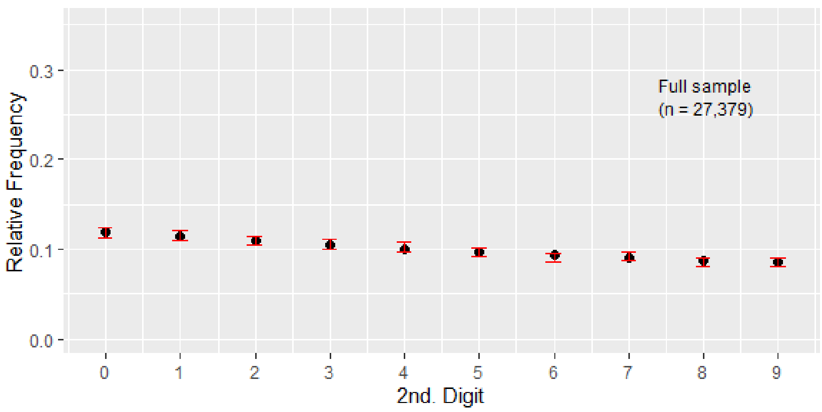


Figure 7(a): 95% Goodman Confidence Intervals
for Std. Error 1st. Digits

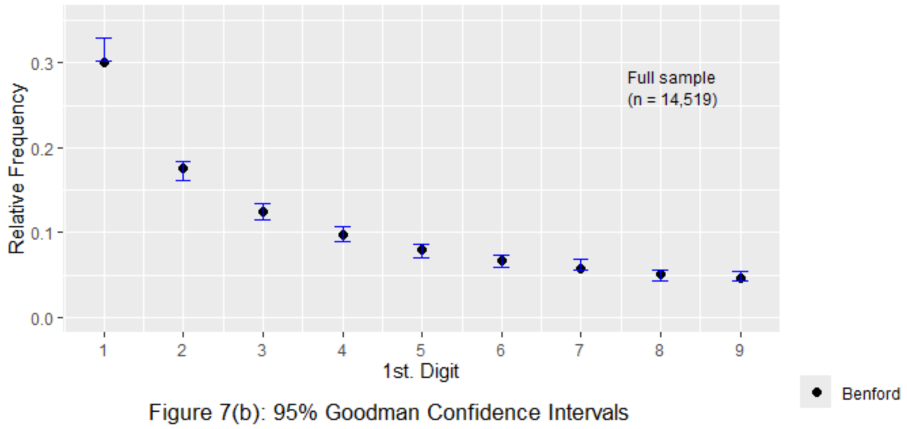


Figure 7(b): 95% Goodman Confidence Intervals
for Std. Error 2nd. Digits

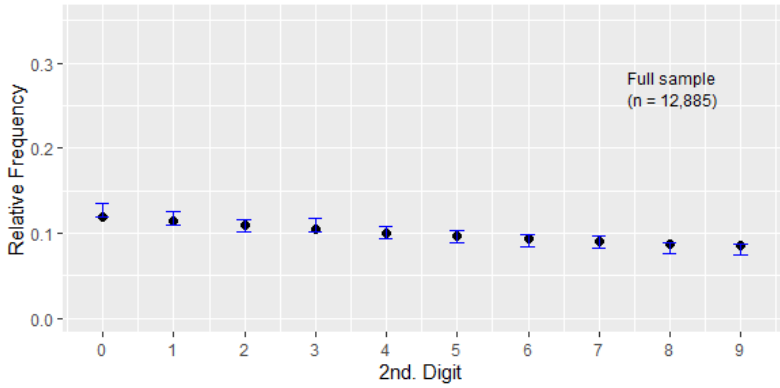


Figure 8(a): 95% Goodman Confidence Intervals
for t-Stat. 1st. Digits

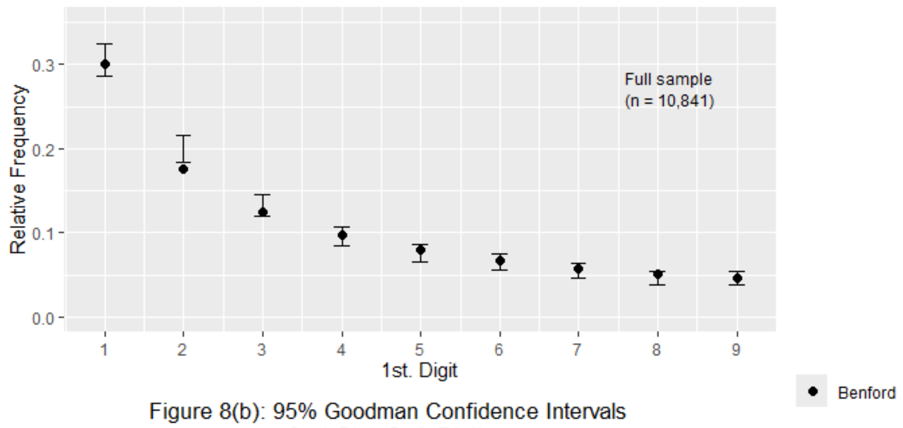
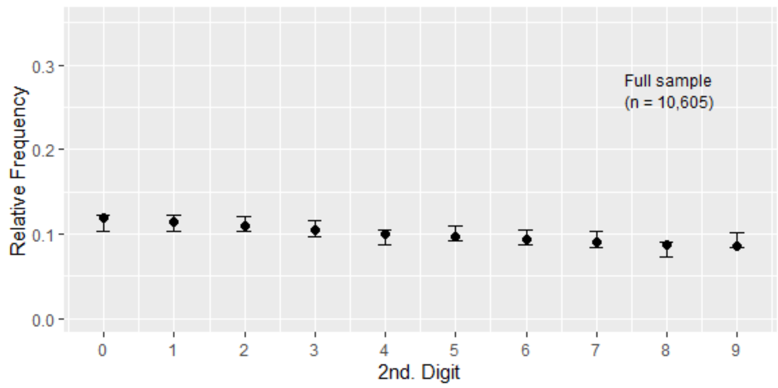


Figure 8(b): 95% Goodman Confidence Intervals
for t-Stat. 2nd. Digits



The full-sample results of the various frequentist tests appear in Table 5. As is explained in the footnote to that table, p-values are reported for some of the tests and in other cases it is only practical to report critical values that have been computed by other authors. The standard χ^2 test rejects Benford's Law for the first significant digit at the 5% level for each of the coefficient, standard error, and t statistic data; and at the 1% level for the first and third of these. For the second significant digits, the corresponding test rejects Benford's Law for the t-statistics at the 5% level, with a marginal result at this level for the standard error data. The 2nd - digit law cannot be rejected for the coefficient data at any reasonable significance level. Moreover, when the χ^2 tests are applied with the modified null asymptotic distribution suggested by Cerquiti and Lupi (2022), all of the p-values equal 1.0, in both the 1st - digit and 2nd - digit cases, so these are omitted from Table 5. This again supports Benford's Law for the reported regression results.

The results from Freedman's EDF test strongly support Benford's Law, with the only marginal result being the rejection of the 1st - digit hypothesis for the standard error data at the 5% significance level (but not at the 2.5% level). The test based on Nigrini's MAD statistic rejects Benford's Law in all of the 1st - digit cases, as well as for the t statistic data in the 2nd - digit case. In the latter case, the result is marginal for the standard error data, while this test supports the 2nd - digit Benford distribution for the coefficient data. Essentially the same conclusions emerge from the results associated with the test based on the MAD test proposed by Kösssler *et al*. (2024).

The *M* test results are mixed. They favour Benford's Law in the 2nd digit cases, and for the 1st digits of the standard error data; but the 1st - digit outcomes for the coefficient and t statistic numbers are negative. Finally, both variants of the invariant sum test reject Benford's Law in all of the cases considered in Table 5. (The p-values for the $IS_{1,M}$ and $IS_{2,M}$ tests are omitted as they are essentially zero.) As might be expected from the earlier discussion of the behaviour of these frequentist tests in very large samples, these results are rather ambiguous. However, the results associated with the Goodman confidence intervals are encouraging. Moreover, these results provide clear motivation for the application of the Bayesian testing under consideration.

Table 6 provides the results associated with the latter when the prior odds between H_0 and H_1 are 1.0. That is, the prior probability of each hypothesis is 0.5. Following Jeffreys (1935), Kass and Raftery (1995, p.777) describe the evidence[8] in favour of H_0 as 'decisive' if $log_{10}(B_{01}) > 2$, 'strong' if $1 < log_{10}(B_{01}) \leq 2$, 'substantial' if $0.5 < log_{10}(B_{01}) \leq 1$, and 'not worth more than a mention' if $0 < log_{10}(B_{01}) \leq 0.5$. In all but one case, the reported Bayes factors provide 'decisive' support for H_0 (Benford's Law). The remaining result, namely $log_{10}(B_{01}) = 0.3566$, is specific to a particular choice of the parameter vector (α_3) for the prior distribution for the probabilities. The three other choices for

Table 5: Frequentist test results - full sample

Test	Coefficients data	Std. error data	t statistic data
		1st Digit	
U_n^{*2}	0.0084	0.1630***	0.0004
MAD_K	4.7157*	3.8509*	5.2734*
$IS_{1,M}$	704.1482*	571.7294*	61.1506*
$IS_{1,E}$	53.1207*	23.9582*	54.4872*
MAD_N	0.0030	0.0036	0.0056
	[$1x10^{-6}$]	[0.0016]	[$4x10^{-8}$]
$\chi^2_{(8)}$	30.6271	17.1414	48.9967
	[0.0002]	[0.0287]	[$6x10^{-8}$]
M	10.3956*	0.8938	19.1730*
		2nd Digit	
U_n^{*2}	0.0073	0.0549	0.0433
MAD_K	2.4368	3.4070***	3.9791*
$IS_{2,M}$	278.9042	197.2511	74.3273
$IS_{2,E}$	300.1859*	202.9029*	64.5208*
MAD_N	0.0015	0.0030	0.0039
	[0.4682]	[0.0446]	[0.0040]
$\chi^2_{(9)}$	8.9283	17.1725	20.5485
	[0.4439]	[0.0461]	[0.0148]
M	0.0828	0.0390	0.0321

Note: p-values appear in square brackets for the MAD_N and $\chi^2_{(k-1)}$ tests. For the U_n^{*2} test the (10%, 5%, 2.5%, 1%) critical values are (0.143, 0.179, 0.215, 0.263) for the 1st digit case, and (0.154, 0.190, 0.226, 0.274) for the 2nd digit. The (10%, 5%, 1%) critical values for MAD_K are (2.869, 3.084, 3.485) and (3.18, 3.42, 3.92) respectively for the 1st and 2nd digit cases. The (10%, 5%, 1%) critical values for $IS_{1,E}$ and $IS_{2,E}$ tests are (15.038, 17.462, 22.644) and (26.067, 30.036, 38.463) respectively. * and ** denote significance at the 1% and 5% levels respectively. All critical values are asymptotic.

this prior yield decisive results. Two specific results relating to Benford's Law for the first significant digits of the t statistic data are flagged in Table 6 because the associated posterior probability of H_0 falls below 1.0000. However, one of these results is still above 99% and the other exceeds 86% in favour of H_0. Moreover, as the table's footnote explains, even these results are sensitive only to extreme changes in the prior odds. These two results are associated with the first digits of the t-statistic data, consistent with the confidence intervals in Figure 8(a) failing to provide 95% coverage.

Table 6: Bayes factors and posterior probabilities – full sample

	Coefficient numbers		**Std. error numbers**		**t-statistics numbers**	
	$log_{10}(B_{01})$	$p(H_0\|x,\alpha)$	$log_{10}(B_{01})$	$p(H_0\|x,\alpha)$	$log_{10}(B_{01})$	$p(H_0\|x,\alpha)$
			1st Digit			
α_1	8.1109	1.0000	9.8105	1.0000	2.5044	0.9969*
α_2	12.8304	1.0000	14.4713	1.0000	7.0829	1.0000
α_3	6.1029	1.0000	7.7386	1.0000	0.3566	0.8688**
α_4	13.4270	1.0000	15.0606	1.0000	7.6627	1.0000
			2nd Digit			
α_1	11.5032	1.0000	9.5288	1.0000	8.5782	1.0000
α_2	17.8200	1.0000	15.8299	1.0000	14.8986	1.0000
α_3	11.0813	1.0000	9.0931	1.0000	8.1659	1.0000
α_4	17.8416	1.0000	15.8500	1.0000	14.9210	1.0000

Note: * and **. If the prior odds are changed to 1 in 3 (1 in 9) in favour of Benford's Law, these two posterior probabilities *in favour* of that law fall to 0.9907 (0.9726) and 0.4310 (0.2016) respectively. None of the other tabulated results change.

In short, the Bayesian testing provides 'decisive' and robust evidence that supports Benford's Law, for both the first and second significant digits, for each of the full samples of coefficient, standard error, and t statistic results reported in *NZEP*. The clarity of this result stands in sharp contrast to the ambiguity of the results from the frequentist tests. This is not surprising.[9]

5.2 Sub-sample results

There are three reasons for extending our analysis to deal with certain sub-samples of our data. The first is to investigate if there is any change in the conformity with Benford's Law over the history of publication of *NZEP*. Accordingly, we consider the six sub-samples associated with the decades of publication, the details of which are summarized in Table 7. This also accords with our second reason for sub-sample analysis, as it provides results that can be used to assess the robustness of our full-sample results and conclusions. In addition, the smaller (and more variable) sample sizes shown in Table 7 largely avoid some of the potential issues associated with 'very large *n*' that received attention in section 4.1. The robustness of our results, and the effects of more modest sample sizes on our conclusions regarding accordance with Benford's Law are also examined by applying the various statistical tests to samples of sizes 500, 1,000 and 2,000 drawn randomly (without replacement) from the three groups of data in the full sample.

5.2.1 Decade sub-samples

The results based on the decade sub-samples appear in Table 7. The relative frequencies of first digits for the coefficient, standard error, and t statistic data are compared with the corresponding Benford probabilities in Figures (a) to 9(f). Figures 10(a) to 10(f) provide the same information in the 2nd - digit case.

Table 7: Decade sub-sample sizes

			1st Digits			
	1966-1979	**1980-1989**	**1990-1999**	**2000-2009**	**2010-2019**	**2020-2025**
Coef	3,013	950	2,996	4,264	11,744	7,192
s.e.	1,258	449	910	2,487	5,224	4,191
t-stat	1,232	261	1,685	1,402	5,225	1,036
			2nd Digits			
	1966-1979	**1980-1989**	**1990-1999**	**2000-2009**	**2010-2019**	**2020-2025**
Coef	2,890	894	2,740	3,952	10,661	6,242
s.e.	1,177	389	864	2,212	4,710	3,533
t-stat	1,211	250	1,656	1,377	5,091	1,020

Figure 9 (a): 1966-1979
Relative Frequency of 1st. Digits of Coefficients, Standard Errors, & t-Statistics

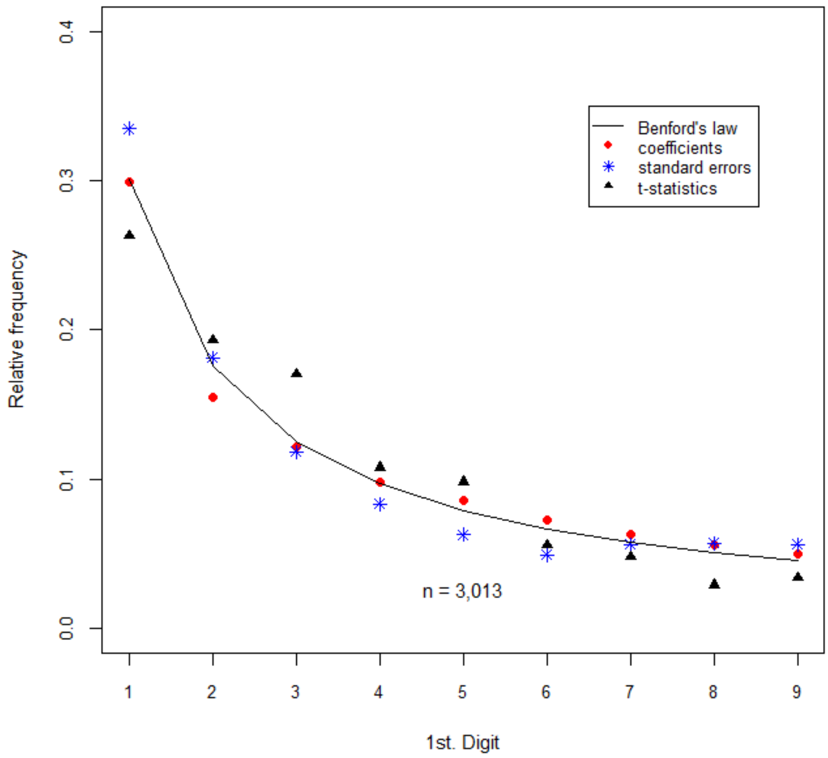


Figure 9 (b): 1980-1989
Relative Frequency of 1st. Digits of Coefficients, Standard Errors, & t-Statistics

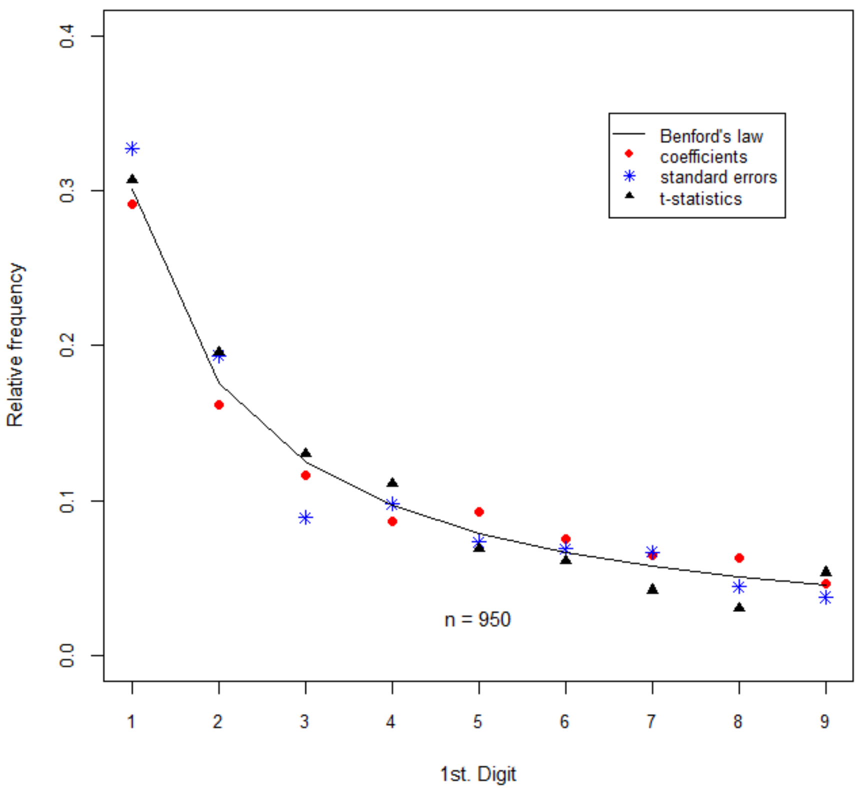


Figure 9 (c): 1990-1999
Relative Frequency of 1st. Digits of Coefficients, Standard Errors, & t-Statistics

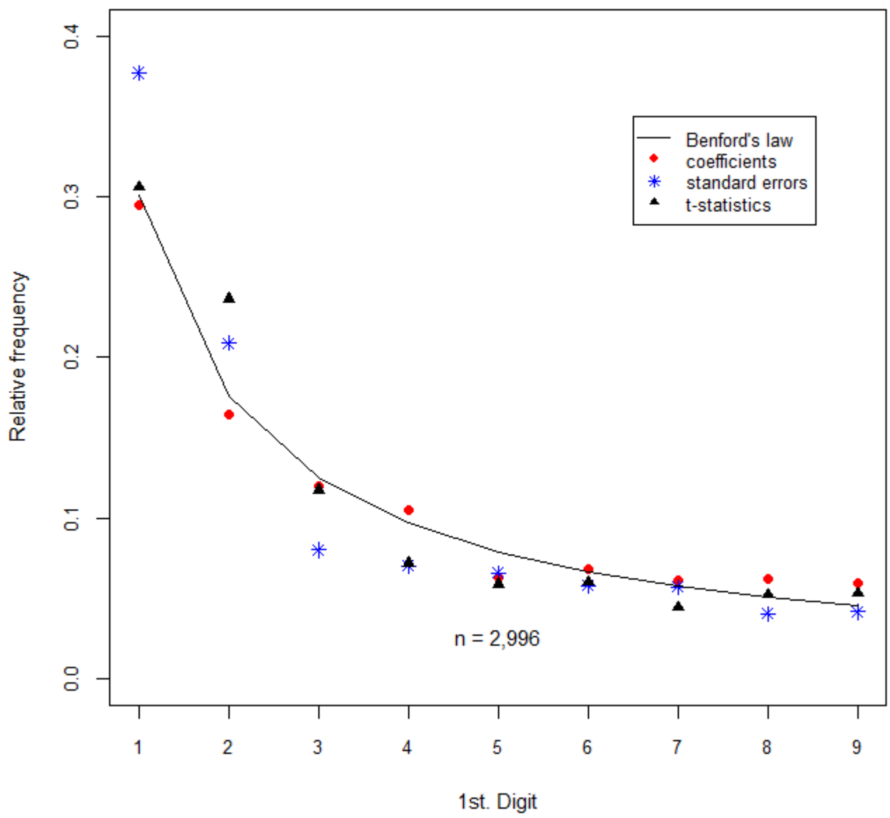


Figure 9 (d): 2000-2009
Relative Frequency of 1st. Digits of Coefficients, Standard Errors, & t-Statistics

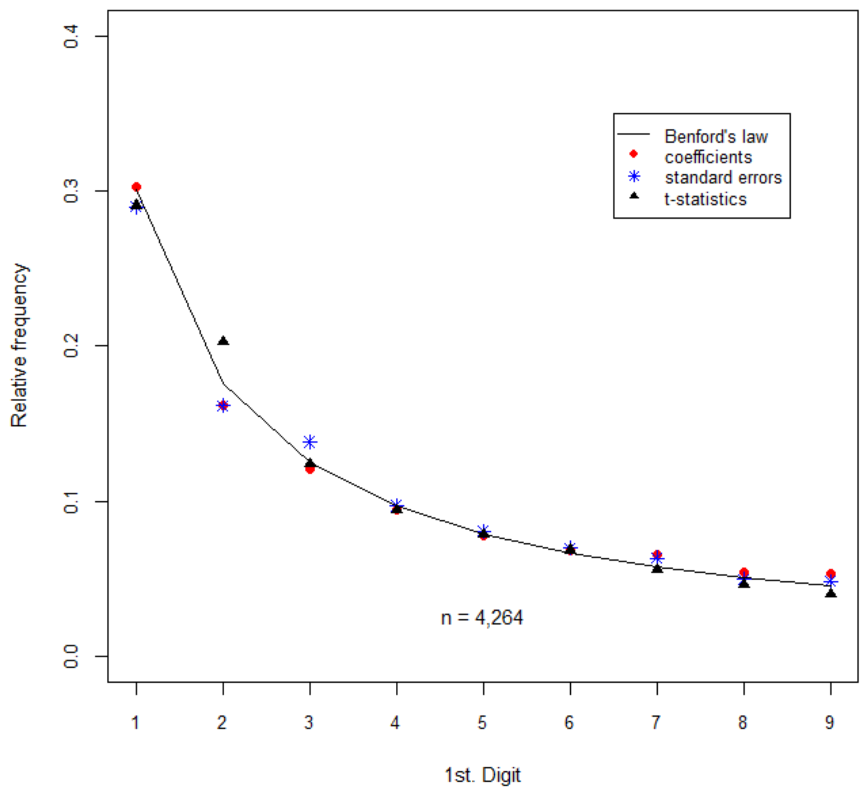


Figure 9 (e): 2010-2019
Relative Frequency of 1st. Digits of Coefficients, Standard Errors, & t-Statistics

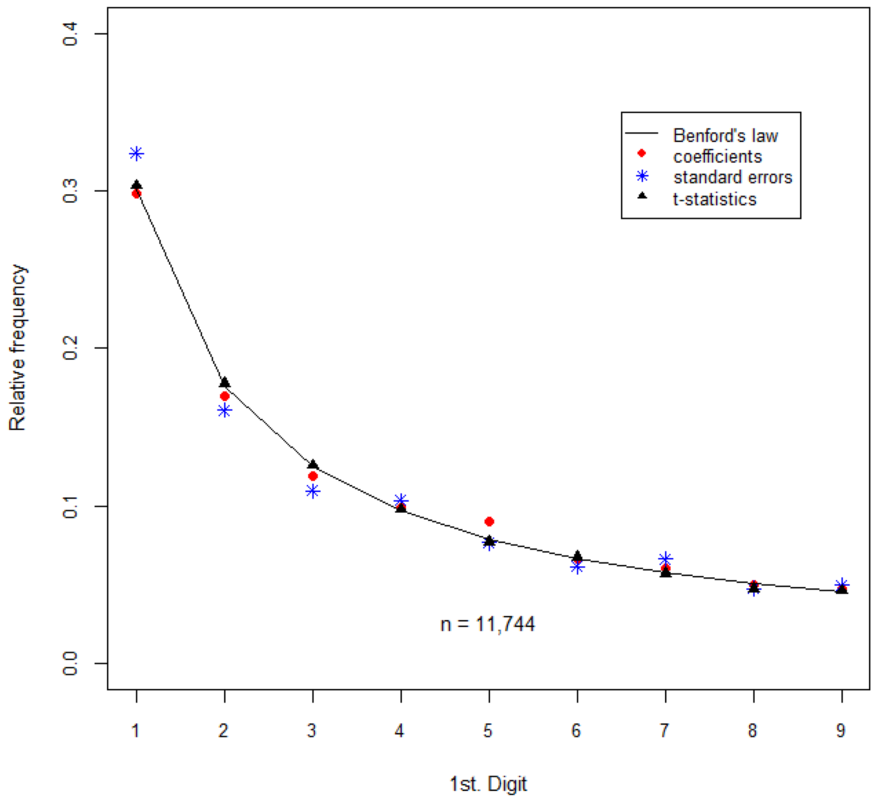


Figure 9 (f): 2020-2025
Relative Frequency of 1st. Digits of Coefficients, Standard Errors, & t-Statistics

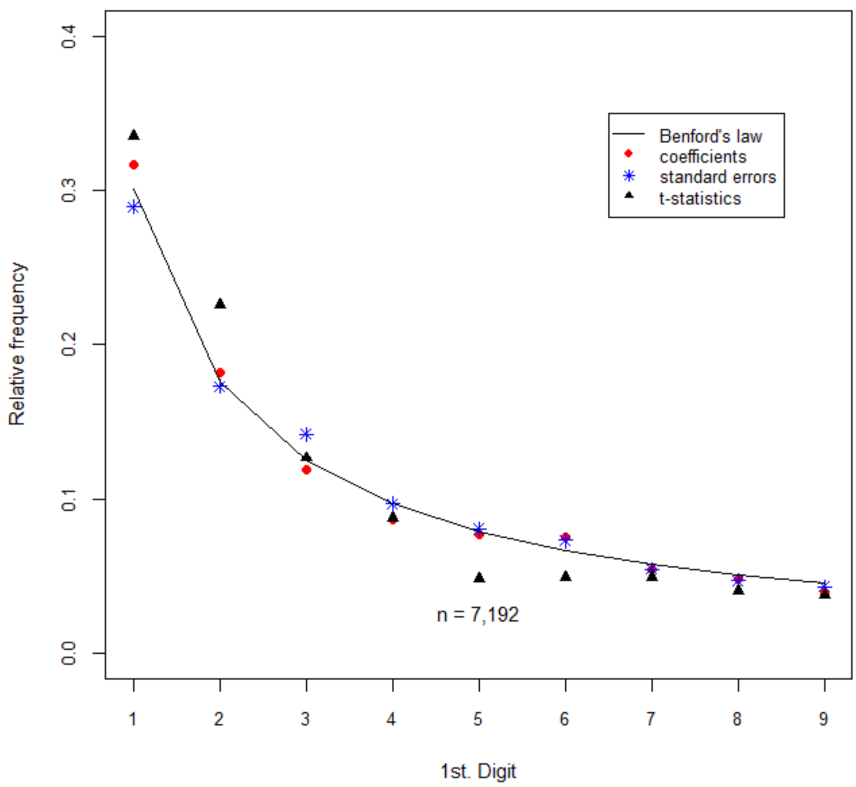


Figure 10 (a): 1966-1979
Relative Frequency of 2nd. Digits of Coefficients, Standard Errors, & t-Statistics

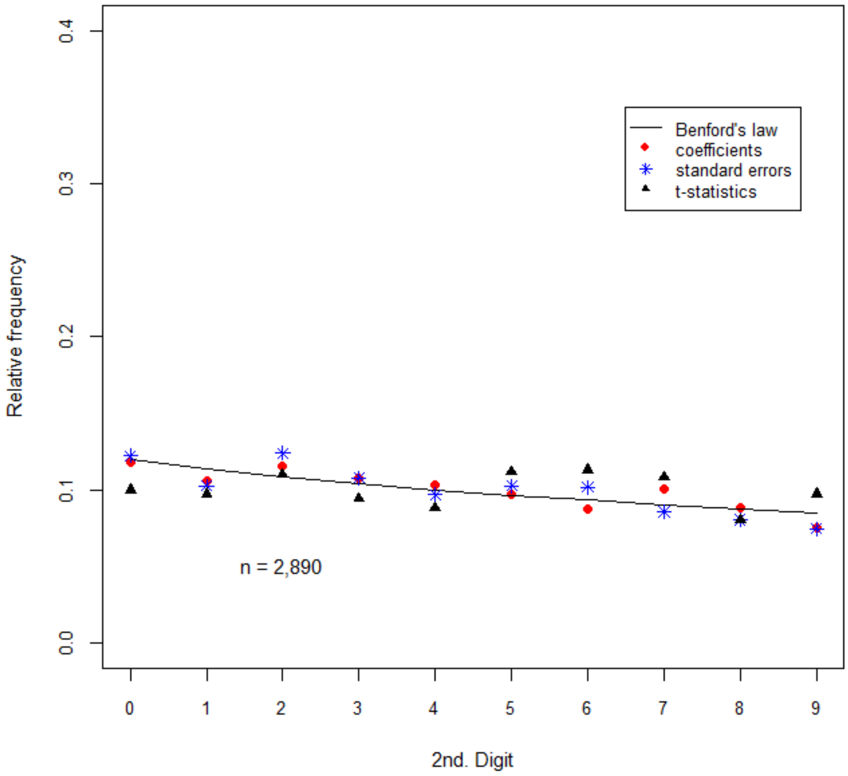


Figure 10 (b): 1980-1989
Relative Frequency of 2nd. Digits of Coefficients, Standard Errors, & t-Statistics

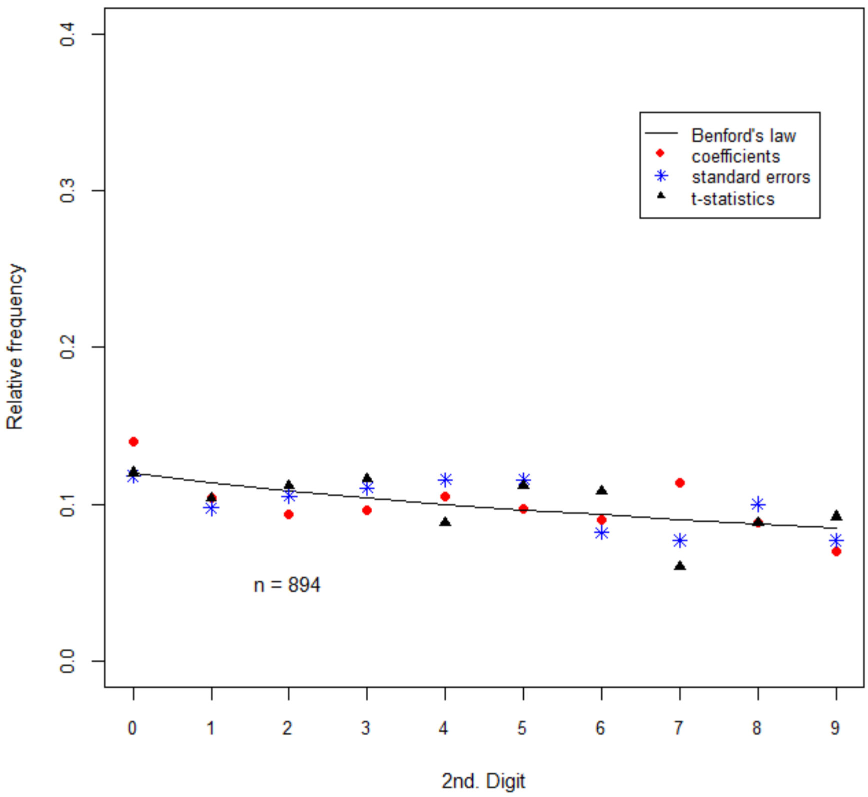


Figure 10 (c): 1990-1999
Relative Frequency of 2nd. Digits of Coefficients,
Standard Errors, & t-Statistics

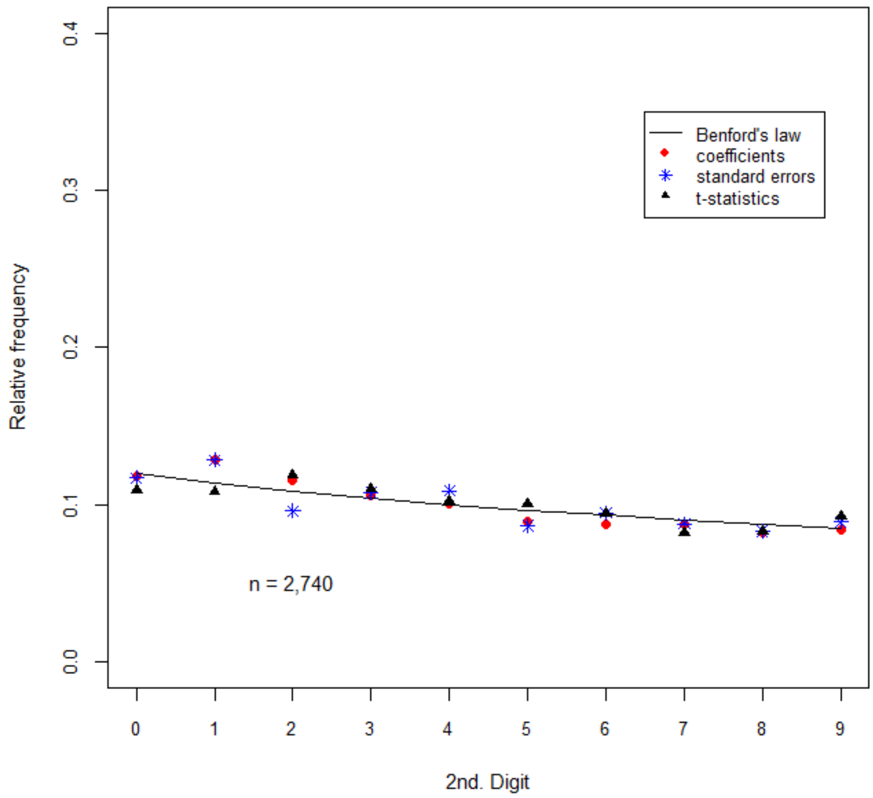


Figure 10 (d): 2000-2009
Relative Frequency of 2nd. Digits of Coefficients,
Standard Errors, & t-Statistics

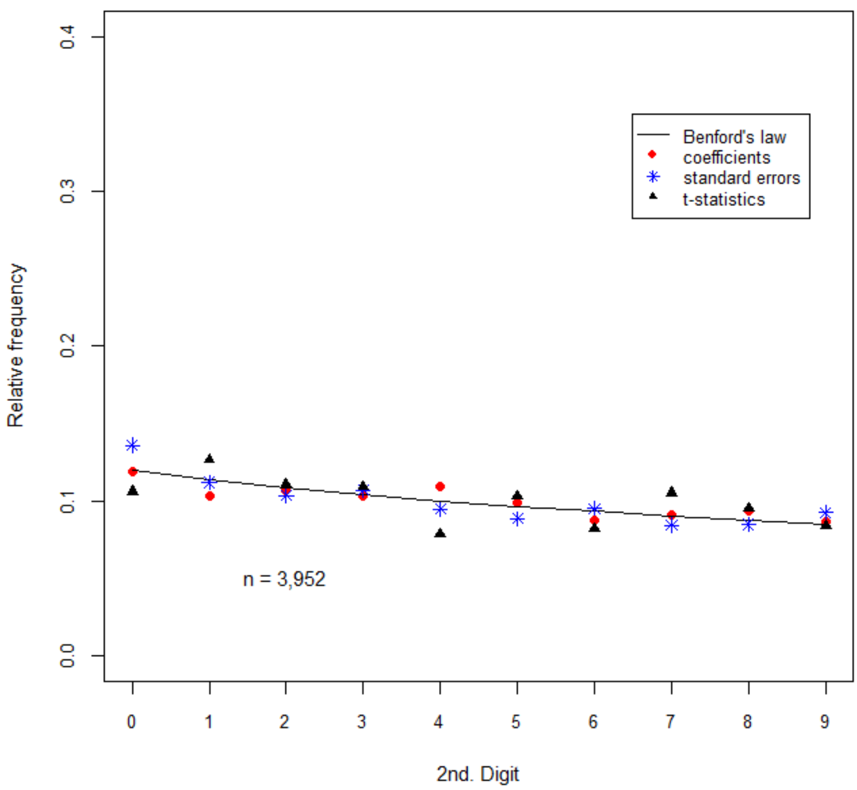


Figure 10 (e): 2010-2019
Relative Frequency of 2nd. Digits of Coefficients, Standard Errors, & t-Statistics

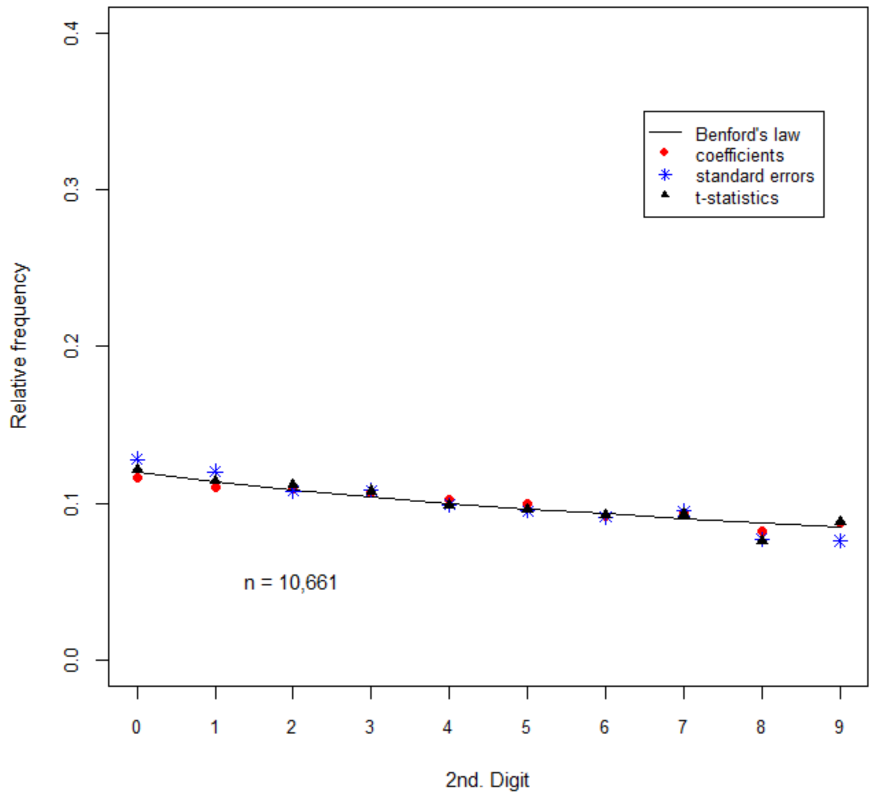
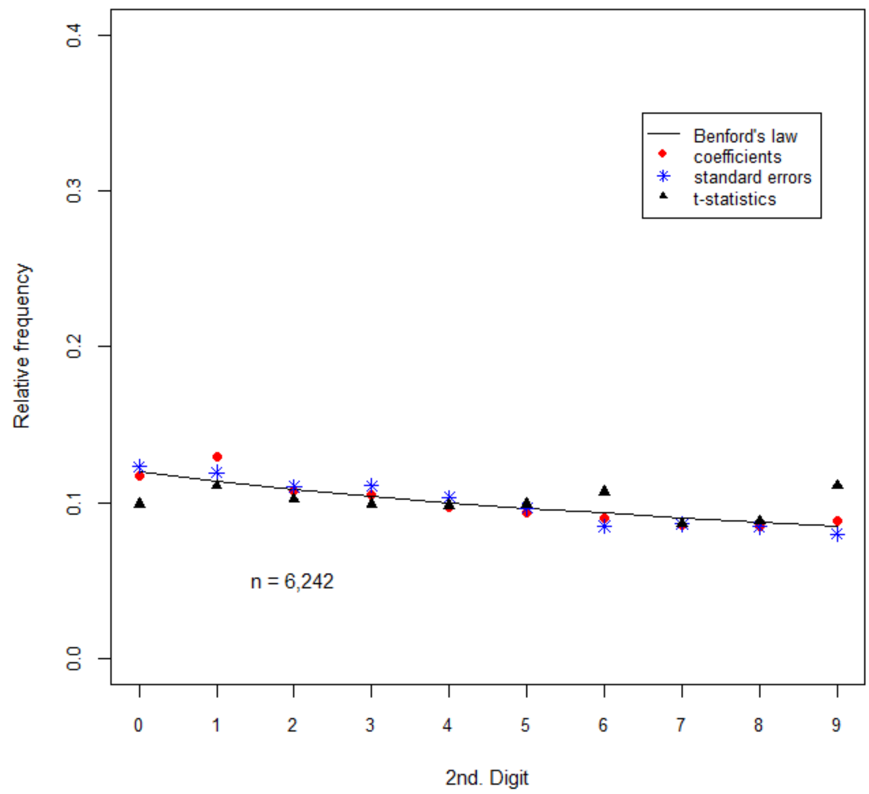


Figure 10 (f): 2020-2025
Relative Frequency of 2nd. Digits of Coefficients, Standard Errors, & t-Statistics



The outcome of the Bayesian analysis is summarized in Table 8, where only the results associated with the 'least informative' natural conjugate prior for the parameters (the probabilities) are presented. This corresponds to the choice, $\alpha_3 = (cp_1, cp_2, \dots\dots, cp_k\ )$, where c = 22 when H_0 is the 1st - digit Benford distribution, c = 12 when H_0 is the 2nd - digit Benford distribution, and the p_i's are the associated Benford probabilities. This presentational choice is made because for this α the smallest $log_{10}(B_{01})$ values and $p(H_0|x, \alpha\ )$ values are obtained for the coefficient, standard error, and t statistic data, for both the 1st - digit and 2nd - digit hypotheses. The same situation holds for the results associated with α_3 in Table 6. So, the results reported in Table 8 are those that provide the *least* support for Benford's Law, across all of the situations considered.

However, this support is very strong, with the posterior probabilities in favour of Benford's Law all equal to unity for the 2nd - digit analysis. Without exception, the results in Table 8 provide 'decisive' support for the 2nd - digit Benford distribution. This support spans the decades of development of the *NZEP*, and it is based on samples that range in size from n = 251 (for the t-statistics reported in 1980-1989), to n = 10,661 for the regression coefficients reported in 2010-2019. In short, this illustrates the robustness of the 2nd - digit results for the full sample in Table 6. The Bayesian test results in Table 8 for the 1st - digit Benford distribution exhibit some variation across the sub-samples. However, all of the posterior probabilities in favour of Benford's Law for the coefficient second digits exceed 90%, all but one exceed 90% for the standard error data, and three of the six are essentially unity in value for the t-statistics. The other three results for the t statistic first digits strongly favour the alternative hypothesis that Benford's Law *does not hold*. Of the eighteen 1st - digit Bayes factors, twelve provide support for Benford's Law that is 'decisive', and two more provide 'substantive' support. Among the sub-samples, the 1st - digit evidence in favour of the 1st - digit Benford distribution is least for 1990-1999, and strongest for 1980-1989 and 2000-2009. If the focus is on the digits of the reported coefficient estimates, the Benford's Law is supported as much for each decade of data as it was for the full sample. The sensitivity of the results to the timing of the data lies essentially with the standard errors and t-statistics.

Some illustrative results based on some of the frequentist tests for the decade sub-sample data are also provided. Table 9 presents the results of Freedman's U_n^{*2} test. As was the case for the full sample results in Table 5, these test results overwhelmingly support Benford's Law at both the 1st - digit and 2nd - digit levels. Interestingly, some of the rejections of H_0 in Table 9 are consistent with their counterparts in Table 8. These are for the first digits of the standard errors and t-statistics in 1990-1999 and 1966-1979 respectively; and to a lesser degree for the standard error first digits in 2010-2019. The use of the χ^2 GOF test is still questionable for some of the sub-sample sizes. However, given its traditional usage, we report the results of this test for the decade sub-samples in Table 10.

Table 8: Bayes factors and posterior probabilities – decade sub-samples

	Coefficient data		Std. error data		t statistic data	
	$log_{10}(B_{01})$	$p(H_0\|x,\alpha)$	$log_{10}(B_{01})$	$p(H_0\|x,\alpha)$	$log_{10}(B_{01})$	$p(H_0\|x,\alpha)$
			1st Digit			
1966-1979	5.5029	1.0000	2.3643	0.9957	-4.7880	1×10^{-5}
1980-1989	4.8165	1.0000	3.6091	0.9997	3.3505	0.9996
1990-1999	0.9961	0.9084	-4.3380	5×10^{-5}	-5.8808	1×10^{-6}
2000-2009	5.8693	1.0000	6.3495	1.0000	5.6898	1.0000
2010-2019	5.5210	1.0000	1.0858	0.9242	9.2386	1.0000
2020-2025	3.0538	0.9991	5.6795	1.0000	-2.5740	0.0027
			2nd Digit			
1966-1979	15.0680	1.0000	14.2015	1.0000	10.6746	1.0000
1980-1989	12.3621	1.0000	12.5648	1.0000	11.8350	1.0000
1990-1999	15.2569	1.0000	14.2405	1.0000	14.9910	1.0000
2000-2009	15.5711	1.0000	14.8176	1.0000	12.2926	1.0000
2010-2019	17.2782	1.0000	14.5584	1.0000	16.1550	1.0000
2020-2025	14.7983	1.0000	16.0834	1.0000	12.5857	1.0000

The p-values based on the non-central χ^2 critical values suggested by Cerqueti and Lupi (2022) all take the value unity, as was the case in Table 5, so they are not included here. The implications of the 2nd - digit results in Table 10 closely match those in Tables 8 and 9. With only one exception, relating to the 1966-1979 t-statistics, they support Benford's Law at the 2.5% significance level. Again, the 1st - digit results are mixed. Moreover, the four such results that were significant in Table 9 are among the ten that are significant at the 2.5% level in Table 10.

The 95% Goodman confidence intervals for the various results based on the 1990-1999 sub-sample appear in Figures 11 to 13. This sub-sample was chosen for illustrative purposes because in this decade Benford's 1st - digit distribution has the least support, based on the results in Tables 8 to 10. The confidence intervals for the other decades have even better coverage of the Benford probabilities than these illustrative ones.[10] In Figure 11(a), the confidence intervals cover the probabilities, which is consistent with the U_n^{*2} test result in Table 9, and the Bayes factor result in Table 8. However, the M test and χ^2 test results in Tables 9 and 10 conflict with this. The confidence intervals' coverage in Figures 12(a) and 13(a) accord almost exactly with the Bayesian and frequentist test results in Tables 8 to 10. The confidence intervals' coverage for the 2nd - digit probabilities in all cases matches all of the test results at the 5% level. In short, there is strong support for Benford's Law for the decade sub-samples, and the Goodman confidence intervals provide a valuable frequentist complement to the Bayesian tests.

Table 9: U_n^{*2} and M test results - decade sub-samples

	Coefficient data		**Std. error data**		**t statistic data**	
	U_n^{*2}	M	U_n^{*2}	M	U_n^{*2}	M
			1st Digit			
1966-1979	0.0016	7.9241*	0.1405	1.4337	0.1761***	4.2497*
1980-1989	0.0084	4.4490**	0.0308	1.0548	0.0008	1.2718
1990-1999	0.0105	12.6310*	0.5177*	17.7061*	0.0035	7.0059*
2000-2009	0.0007	7.8350*	0.0304	2.8094***	0.0159	1.2223
2010-2019	0.0086	4.1929**	0.2651*	0.0097	0.0023	0.6017
2020-2025	0.1753***	9.2412*	0.0580	0.0002	0.1177	24.4681*
			2nd Digit			
1966-1979	0.0019	0.0343	0.0179	0.0140	0.0042	0.0144
1980-1989	0.0121	0.8656	0.0120	0.0125	0.0 005	0.0080
1990-1999	0.0360	3.7309***	0.0062	1.1765	0.0229	2.2549
2000-2009	0.0362	2.3760	0.0004	1.3299	0.0306	0.8279
2010-2019	0.2157**	0.6663	0.0078	0.2943	0.0007	0.3182
2020-2025	0.2079**	2.0344	0.0954	1.1515	0.0460	0.3324

Note: The (10%, 5%, 1%) critical values for U_n^{*2} are (0.143, 0.179, 0.263) for the 1st digit case, and (0.154, 0.190, 0.274) for the 2nd digit. For M, these critical values are (2.706, 3.841, 6.635) for both the 1st digit and 2nd digit cases. Accordingly, *, **, *** denote significance at the 1%, 5%, and 10% levels.

Table 10: $\chi^2_{(k-1)}$ test results - decade sub-samples

	Coefficient data		**Std. error data**		**t statistic data**	
	$\chi^2_{(8)}$	**p-val.**	$\chi^2_{(8)}$	**p-val**	$\chi^2_{(8)}$	**p-val**
			1st Digit			
1966-1979	15.1281	0.0057	21.9112	0.0051	55.2027	4×10^{-9}
1980-1989	9.7653	0.2819	8.2684	0.4077	5.2785	0.7274
1990-1999	37.0387	1×10^{-5}	49.4722	5×10^{-8}	63.9849	8×10^{-11}
2000-2009	16.6142	0.0344	9.8142	0.2783	8.0925	0.4245
2010-2019	26.4599	0.0009	39.6197	4×10^{-6}	1.9851	0.9815
2020-2025	32.7975	7×10^{-5}	17.1823	0.0283	41.9437	1×10^{-6}
	$\chi^2_{(9)}$	**p-val.**	$\chi^2_{(9)}$	**p-val**	$\chi^2_{(9)}$	**p-val**
			2nd Digit			
1966-1979	11.2576	0.2585	7.8548	0.5488	24.6896	0.0033
1980-1989	14.2548	0.1135	5.7681	0.7629	4.8030	0.8511
1990-1999	10.1067	0.3419	5.0209	0.8325	7.1201	0.6246
2000-2009	11.3910	0.2499	10.4591	0.3146	17.5776	0.0404
2010-2019	9.7271	0.3730	16.9893	0.0489	10.1417	0.3391
2020-2025	18.6667	0.0282	8.0640	0.5277	14.6925	0.0997

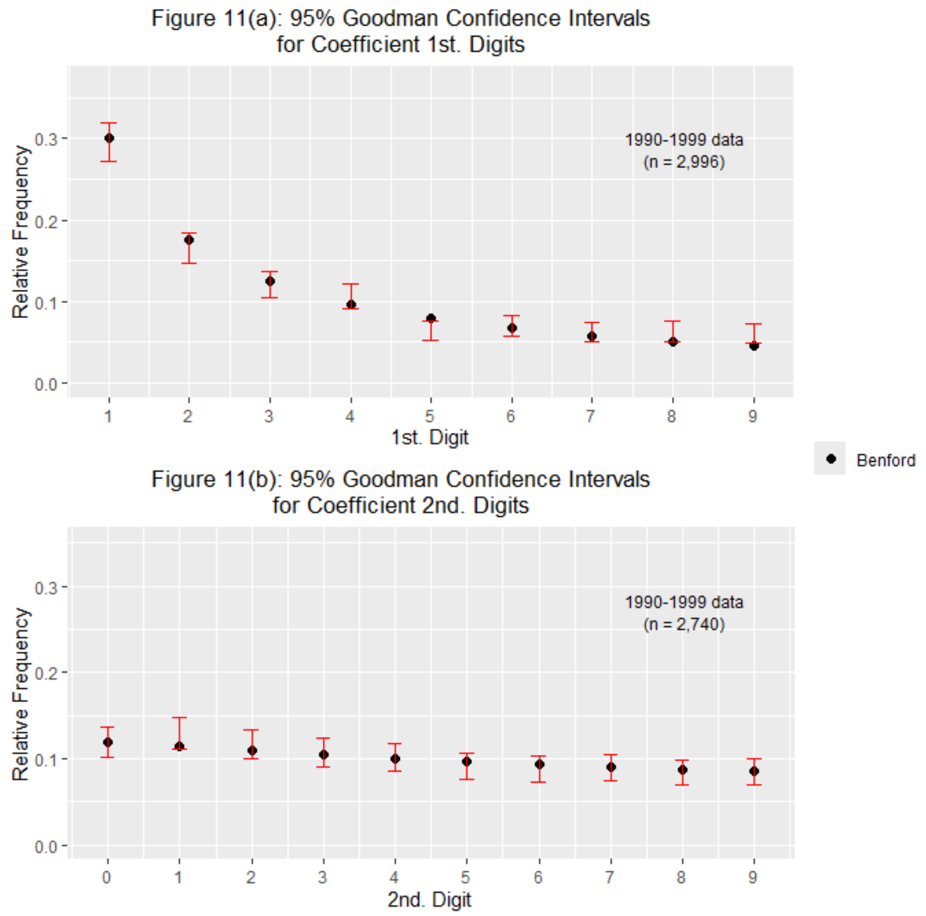


Figure 11(a): 95% Goodman Confidence Intervals for Coefficient 1st. Digits

Figure 11(b): 95% Goodman Confidence Intervals for Coefficient 2nd. Digits

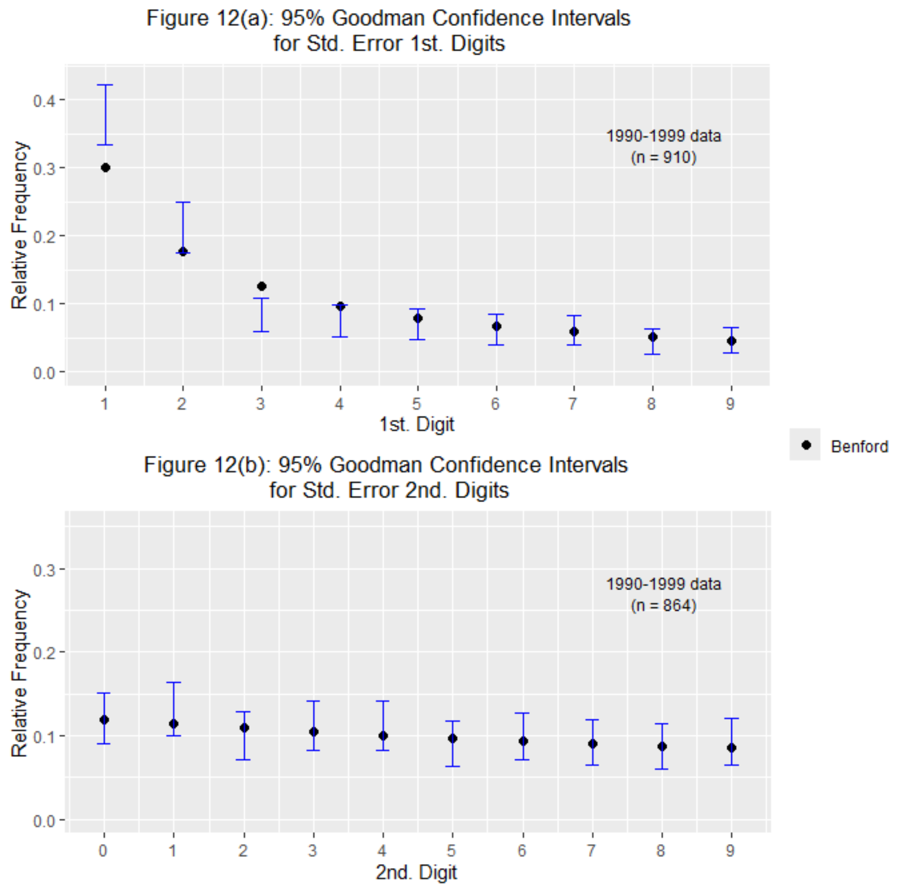
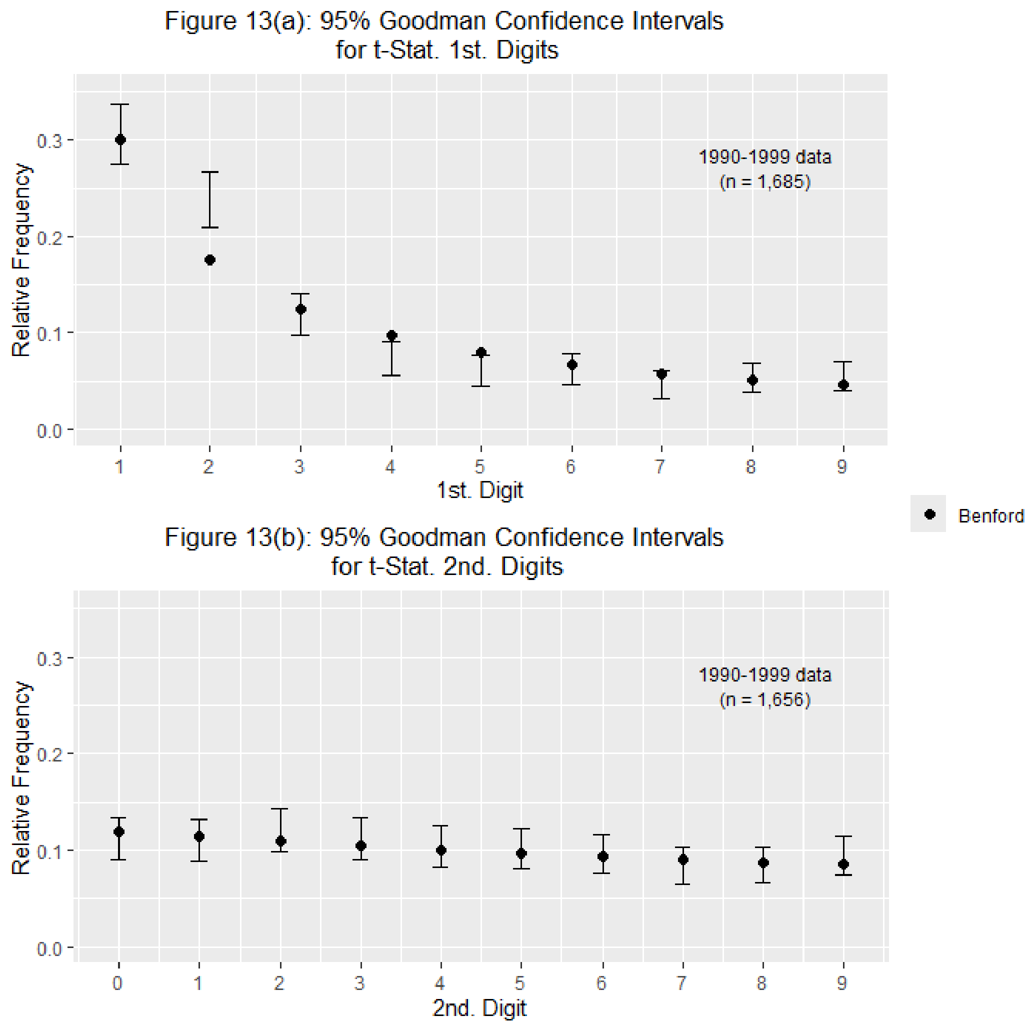


Figure 12(a): 95% Goodman Confidence Intervals for Std. Error 1st. Digits

Figure 12(b): 95% Goodman Confidence Intervals for Std. Error 2nd. Digits



5.2.2 Random sub-samples

The relative frequencies of the first and second digits associated with the random sub-samples of size n = 1,000 appear in Figures 14(a) and 14(b). Similar results[11] were obtained when n = 500 and n = 2,000. Visually, the 1st - digit results appear to be consistent with Benford's law for all three components of the data. The Bayesian analysis was applied to these data, for each random sample. As in section 5.2.1, four different parameters (α) were considered for the prior distribution, and again the results associated with α_3 were the least in favour of Benford's Law. Accordingly, only these results appear in Table 11. These results 'decisively' support Benford's 1st - digit and 2nd - digit laws.

Most of the frequentist test results in Tables 12 and 13 favour Benford's Law for the second digits of all three regression statistics. There are two exceptions to this in the 1st - digit results for both the U_n^{*2} and χ^2 tests, but when the Cerqueti and Lupi (2022) non-central critical values are used to modify the latter test, Benford's Law is fully supported. This support is reinforced by the Goodman confidence intervals in Figures 15 to17, where the 95% intervals cover all of the Benford probabilities[12].

Figure 14 (a): Random sample, n = 1,000
Relative Frequency of 1st. Digits of Coefficients,
Standard Errors, & t-Statistics

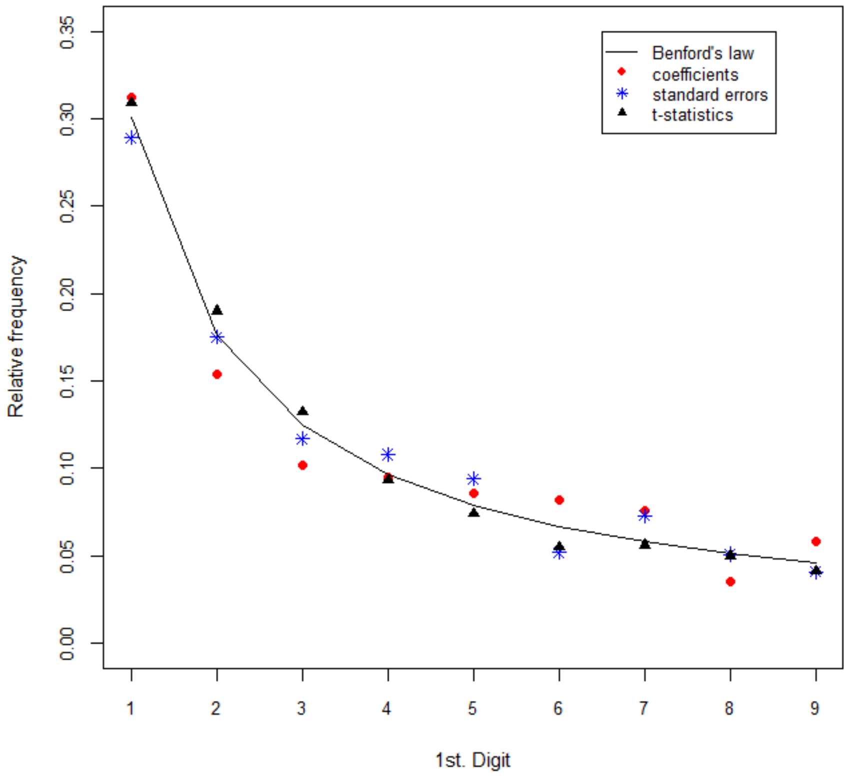


Figure 14 (b): Random sample, n = 1,000
Relative Frequency of 2nd. Digits of Coefficients,
Standard Errors, & t-Statistics

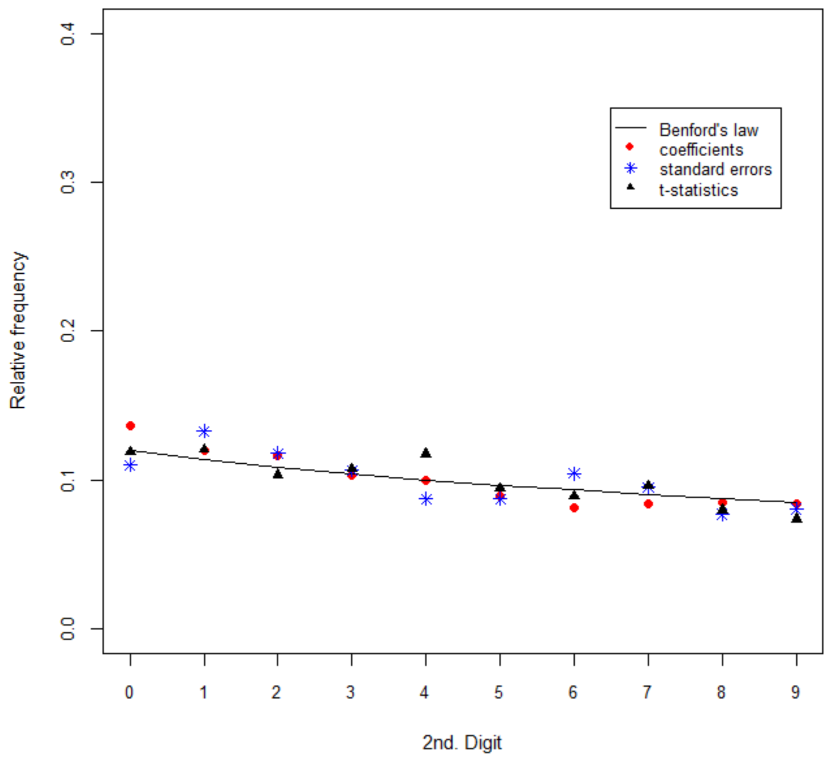


Table 11: Bayes factors and posterior probabilities – random sub-samples

	Coefficient data		**Std. error data**		**t statistic data**	
	$\boldsymbol{log_{10}(B_{01})}$	$\boldsymbol{p(H_0 \mid x,\alpha)}$	$\boldsymbol{log_{10}(B_{01})}$	$\boldsymbol{p(H_0 \mid x,\alpha)}$	$\boldsymbol{log_{10}(B_{01})}$	$\boldsymbol{p(H_0 \mid x,\alpha)}$
$\boldsymbol{n}$			**1st Digit**			
500	5.1554	1.0000	4.7648	1.0000	3.9092	0.9999
1,000	5.7452	1.0000	4.9373	1.0000	3.3294	0.9995
2,000	5.7703	1.0000	5.2876	1.0000	3.3671	0.9996
$\boldsymbol{n}$			**2nd Digit**			
500	4.9542	1.0000	4.1870	0.9999	4.8742	1.0000
1,000	5.3024	1.0000	3.8057	1.0000	5.3027	1.0000
2,000	6.0595	1.0000	6.9185	1.0000	3.8145	0.9998

Table 12: $\boldsymbol{U_n^{*2}}$ and $\boldsymbol{M}$ test results - random sub-samples

	Coefficient data		**Std. error data**		**t statistic data**	
	$\boldsymbol{U_n^{*2}}$	$\boldsymbol{M}$	$\boldsymbol{U_n^{*2}}$	$\boldsymbol{M}$	$\boldsymbol{U_n^{*2}}$	$\boldsymbol{M}$
$\boldsymbol{n}$			**1st Digit**			
500	0.0684	0.1902	0.0117	0.7928	0.3793^{*}	0.4768
1,000	0.0073	2.4573	0.0549	0.4464	0.0433	2.4573
2,000	0.1533^{***}	1.9072	0.1411	0.7928	0.0041	1.4166
$\boldsymbol{n}$			**2nd Digit**			
500	4×10^{-6}	1.2165	0.0008	1.2219	0.0175	1.3248
1,000	0.0012	3.1710^{***}	0.0020	3.0767^{***}	0.0029	3.4155^{***}
2,000	0.0014	1.4695	0.0018	1.4321	0.0069	1.6024

Note: The (10%, 5%, 1%) critical values for U_n^{*2} are (0.143, 0.179, 0.263) for the 1st digit case, and (0.154, 0.190, 0.274) for the 2nd digit. For *M*, these critical values are (2.706, 3.841, 6.635) for both the 1st digit and 2nd digit cases. Accordingly, *, *** denote significance at the 1% and 10% levels.

Table 13: $\chi^2_{(k-1)}$ test results - decade sub-samples

	Coefficient data		**Std. error data**		**t statistic data**	
	$\chi^2_{(8)}$	**p-val.**	$\chi^2_{(8)}$	**p-val**	$\chi^2_{(8)}$	**p-val**
n			**1st Digit**			
500	7.6819	0.4651	13.7722	0.0879	12.6993	0.1226
		[0.9660]		[0.7659]		[0.8144]
1,000	5.0069	0.7568	6.2259	0.6219	16.9419	0.0307
		[0.9999]		[0.996]		[0.9393]
2,000	2.2381	0.9728	15.7110	0.0467	7.2301	0.5120
		[1.0000]		[0.9998]		[1.0000]
	$\chi^2_{(9)}$	**p-val.**	$\chi^2_{(9)}$	**p-val**	$\chi^2_{(9)}$	**p-val**
n			**2nd Digit**			
500	14.6151	0.1021	5.6611	0.7733	12.7493	0.1743
		[0.9641]		[0.9997]		[0.9791]
1,000	12.4749	0.1878	4.6062	0.8672	0.4681	
		[1.0000]		[1.0000]		[1.0000]
2,000	4.6664	0.8624	21.5465	0.0104	4.9505	0.8386
		[1.0000]		[1.0000]		[1.0000]

Note: The figures in square brackets are p-values for the χ^2 test based on the non-central critical values, as suggested by Cerqueti and Lupi (2022).

Figure 15 (a): 95% Goodman Confidence Intervals
for Coefficient 1st. Digits

Figure 15 (b): 95% Goodman Confidence Intervals
for Coefficient 2nd. Digits

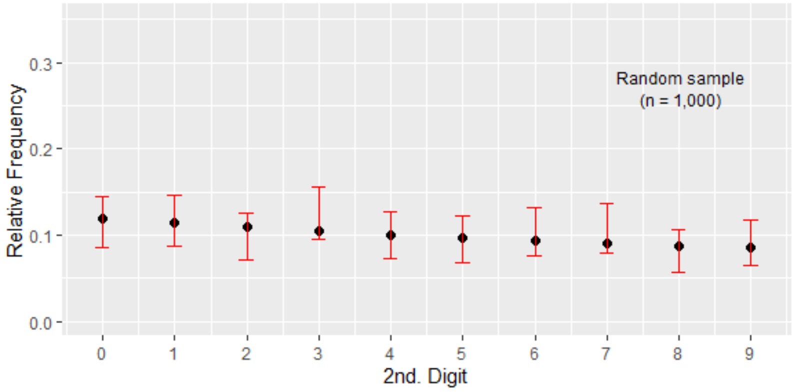
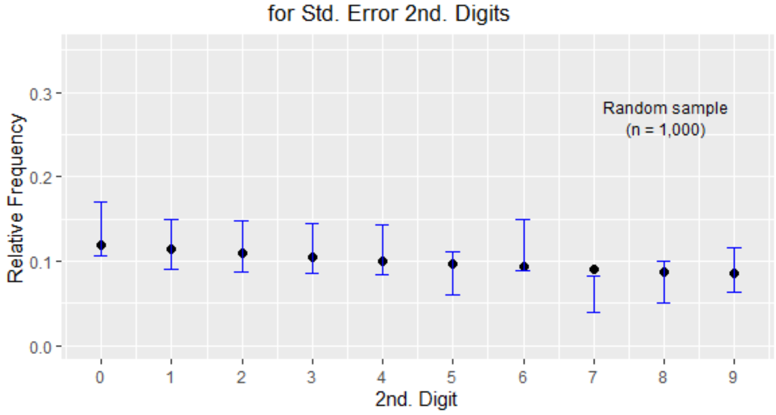
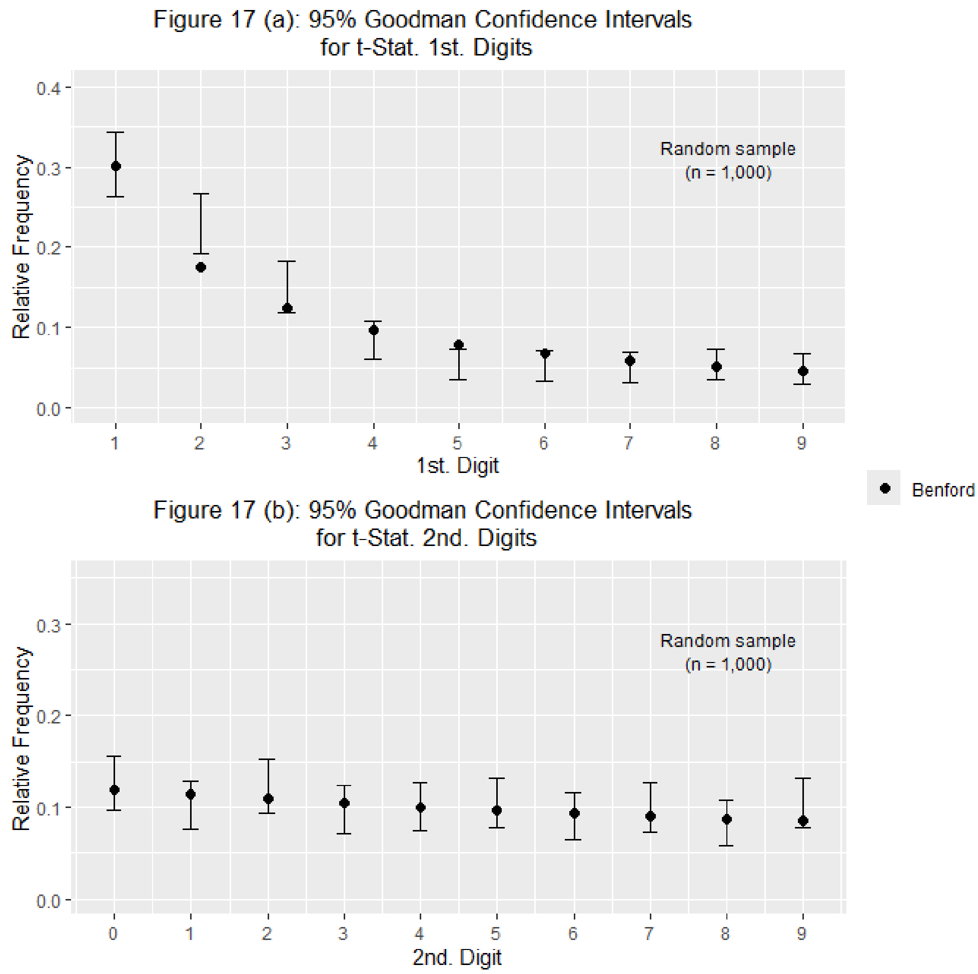


Figure 16 (a): 95% Goodman Confidence Intervals
for Std. Error 1st. Digits

Figure 16 (b): 95% Goodman Confidence Intervals
for Std. Error 2nd. Digits





6. Conclusions and further research

In this paper we have sought to enhance the limited literature that examines the extent to which published regression results in economics journals accord with Benford's Law. To achieve this we used all of the published regression coefficient estimates, standard errors, and Student t-statistics reported in *New Zealand Economic Papers* from its inception in 1966 to the end of 2025. The existing related literature strongly suggests that the first and second significant digits that make up the numbers for such coefficients and standard errors closely follow the 1^{st} - digit and 2^{nd} - digit Benford distributions. Our results agree with this conclusion, and extend it to include regression t-statistics.

Another major addition to this particular literature lies with the statistical procedures that are adopted in this study. In particular, Bayesian methods are used, together with frequentist tests that take account of the very large sample sizes that are involved, and appropriate *joint* confidence intervals. The first and third of these procedures have not been used previously when testing economic regression results for adherence to Benford's Law. Given our analysis of the full sample of data, as well as various sub-samples, we strongly recommend the adoption of Bayesian methods for such studies. In particular, these methods do not suffer from the problems associated with standard frequentist 'goodness of fit' tests when samples are extremely large. The joint (Goodman, 1965) confidence intervals provide a valuable addition to plots of the empirical distributions of the significant digits that are under test, and were found to complement the Bayes factor results very well.

One of the reasons for testing the hypothesis that numbers have significant digits that are consistent with Benford's Law is to check for the possible presence of fraudulent data. Our objective is not to detect fraud in the particular regression studies under examination here. Rather, it is to provide a positive extension to the (limited) existing evidence that such results in the economics literature are sound. Our conclusion is that this is the case. Accordingly, this improves the foundation for subsequent tests of Benford's Law in other regression data where fraudulent activity may be suspected.

There are, however, some caveats relating to this approach to the forensics of regression results. Specifically, Bauer and Gross (2011) conclude that larger samples are needed if the regressions being studied are computed by different analysts, rather than by a single author. They also suggest that Benford's Law will only detect regression fraud if the fraud is extensive. Our study covers many authors, vastly different sets of underlying data, and involves a substantial time-span. The full sample size for our study is large, relative to previous related ones, and the use of sub-samples and multiple tests has established the robustness of our results.

Various extensions of this study could be considered. One obvious possibility is to conduct tests of the *joint* distribution of the first and second significant digits, rather than just the marginal distributions. In addition, the third significant digits could also be tested. Günnel and Tödter (2009) found that the latter extension supported Benford's Law in their study of regression results from the *Empirica* and *Applied Economic*s *Letters* journals. However, as they note, reduced sample sizes are inevitable for the third digits, and this may impact the performance of the tests. An obvious direction for further research is to broaden the Bayesian analysis used in this paper. The basis for our Bayesian testing is the compound multinomial-Dirichlet model. Other choices could be considered, such as the compound binomial-beta model used by Teles da Fonesca (2016). Finally, it would be very beneficial to undertake similar studies with the regression results published in a range of economics journals that focus on different fields within the discipline.

Declaration of interests

The author reports that there are no competing interests to declare.

Acknowledgements

I am very grateful to Mark Holmes, Dorian Owen, Bob Reed, and Dennis Wesselbaum for their very helpful comments.

Notes

1. The database that was created for the present study is available at https://github.com/DaveGiles1949/Data/tree/main/NZEP%20Benford's%20Law%20Data.
2. The 'logarithmic distribution' is, of course, Benford's distribution.
3. Berger *et al*. (2009) provide an online database of these studies. As of 30 May 2026, 2,047 papers, books and websites were listed in the database.
4. Joenssen (2015) provides an R extension package, 'BenfordTests', that includes additional frequentist tests. We found that these tests provide mixed results in samples of modest size, but they suffer from the problems associated with very large samples that were discussed in section 4.1. Accordingly, we do not report results from 'BenfordTests'.
5. Specifically, Roberts's FORTRAN code available at https://www.robertnz.net/download.html was used to evaluate the probabilities of linear combinations of χ^2 random variables.
6. The R code written for the present study is available at https://github.com/DaveGiles1949/r-code/tree/master/NZEP%20Benford'sLaw%20RCode. Version 4.5.2 of the R software was used. The code uses several R extension packages: 'benford.analysis' (Cinelli, 2018),'gridExtra' (Auguie, 2017), 'Matrix' (Bates *et al*., 2026), 'ggpubr' (Kassambara, 2026), 'ggplot2' (Wickham, 2016), 'stringr' (Wickham, 2025), and 'readxl' (Wickham and Bryan, 2025).
7. It was found that the 99% confidence intervals for this case also fail to provide coverage.
8. $log_{10}B_{01}$ values of -0.5, 0, 0.5, 1, and 2 correspond to $Pr(H_0|x, \alpha\ )$ values of 24%, 50%, 76%, 91%, and 99% respectively.
9. An even more dramatic example of this was discussed by Raftery (1986) in the context of a sample of sociological data with n = 113,556. There, a model that explained more than 99% of the sample variation was rejected by a conventional χ^2 test with a p-value of the order of 10^{-120}, but was favoured by the computed Bayes factor which can be shown to be 7.82.
10. The corresponding confidence intervals for the other decade sub-samples are available in the online Appendix A at https://github.com/DaveGiles1949/Results.
11. The relative frequency plots for the random sub-samples with n = 500 and n = 2,000 are available in the online Appendix B at https://github.com/DaveGiles1949/Results.
12. The corresponding confidence intervals based on the random sub-samples with n = 500 and n = 2,000 are available in the online Appendix B at https://github.com/DaveGiles1949/Results.

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